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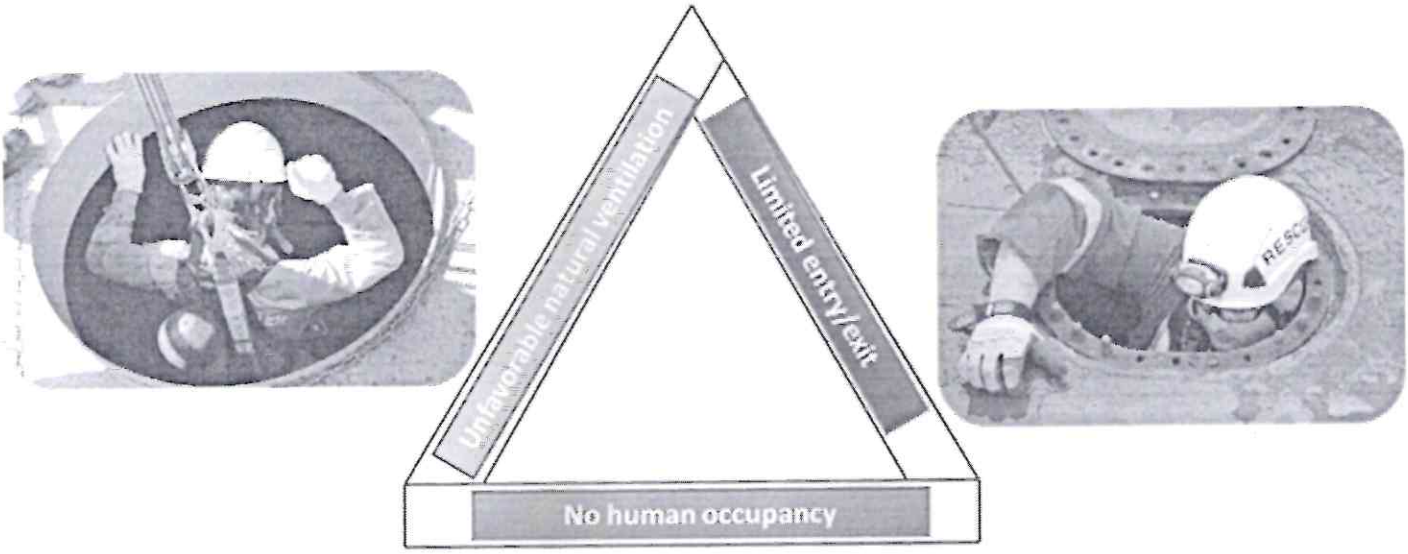
Background:

Confined space safety is the practice of ensuring safe work conditions in fully or partially enclosed areas such as manholes, pipelines, boilers, furnace, tanks, utility vaults and sewers. Confined space safety precautions should be undertaken accordingly to comply with health and safety regulations and prevent work-related injuries, illnesses, or even deaths.



Must have all the THREE elements to be classified as a Confined Space

At Vedanta Industries, Sterlite Copper division, we are involving our employees in Confined space work but have partially sub-contracted the confined space activity to our business partner's/ contract employees.

The various Confined space activities that are undertaken in the premise are as follows:

- Work inside vertical shaft furnace for replacing refractory lining & refurbishment (Routine activity)
- Working in Boilers for repair, maintenance and cleaning purpose (non-routine activity)
- Work inside HSD/ acid tank/ settler tank/ slime tank for repair, maintenance and cleaning purpose (non-routine activity)

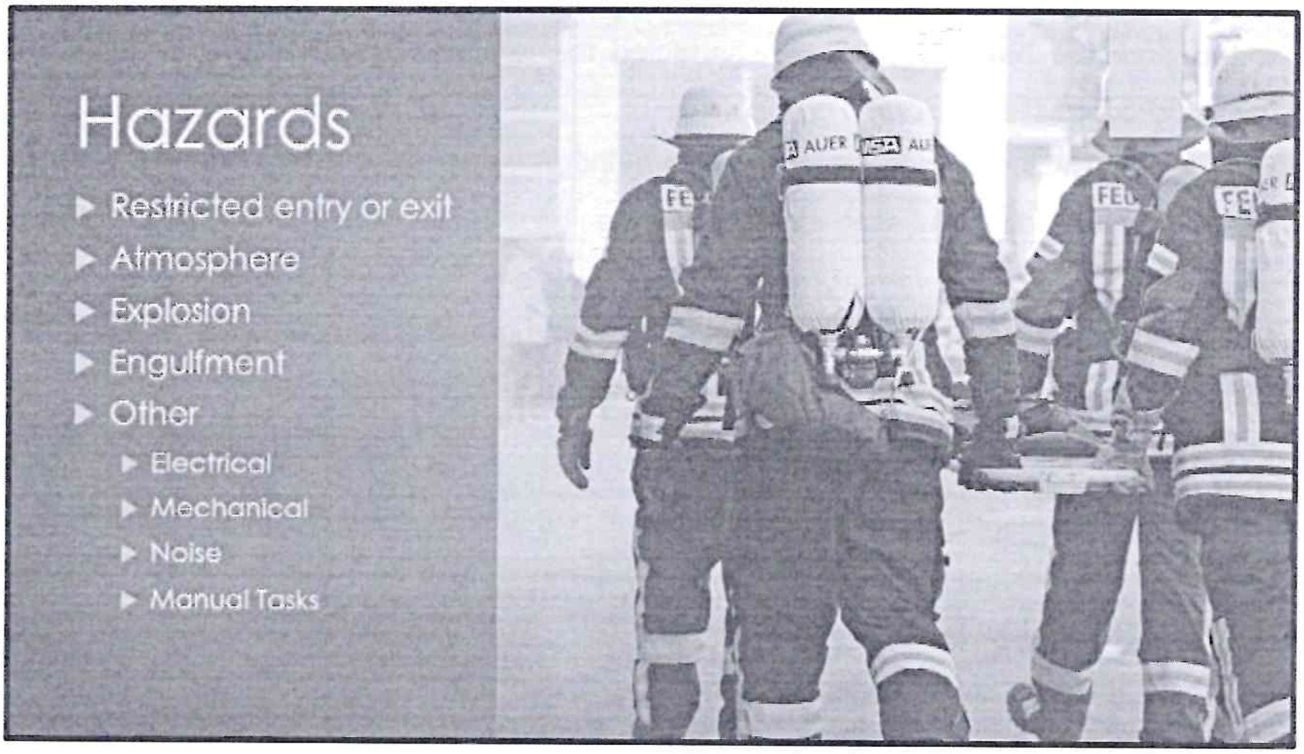
The senior leadership of Vedanta Industries is committed towards health and safety of all stakeholders including business partners and demonstrates by implementing preventive and proactive OH&S management system in the organization by ensuring the following:

- Establishing and maintaining Safe place of work
- Establishing and maintaining Safe plant and equipment
- Hiring Competent people for the task
- Developing and implementing Safe System of work (SSOW)

- Engaging and empowering the workforce through Information, Instruction, Training and Supervision

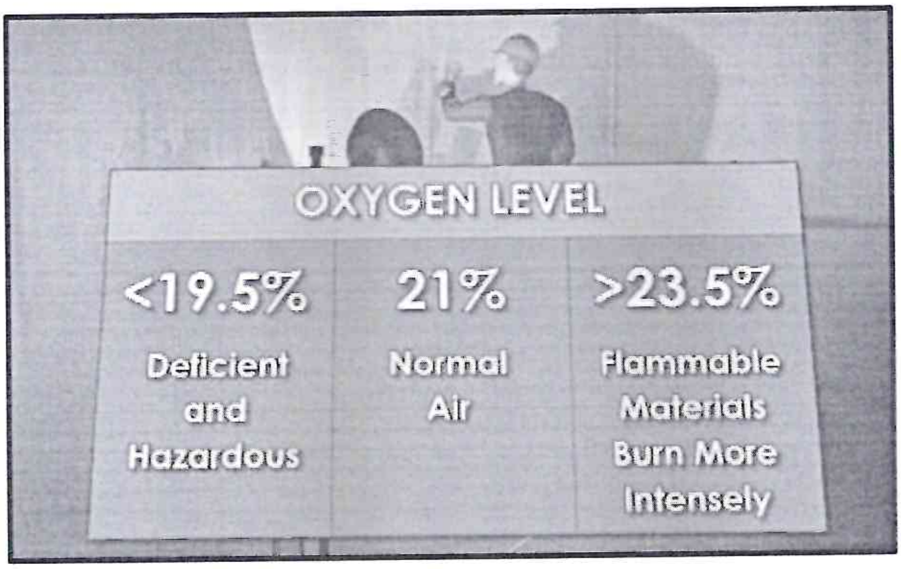
The senior leadership team has assigned the ultimate responsibility of health and safety management to Mr. Leeladhar Patidar, COO. Each employee and business partners are committed and adequately trained in OH&S hazards, risks and control measures. Permit to work procedures are implemented, Inspections and audits are conducted by internal auditors as well as third party auditors to identify lapses in OH&S management system for continual improvement. Adequate Personal Protective Equipment are provided to all employees including business partners.
Line management is responsible for ensuring OH&S of all workforces. HSE organization is established by senior management for developing and sustaining OH&S MS.

Problem area:
Although adequate active and passive safety controls are implemented for safeguarding the business partners and employees engaged for confined space activities, but the senior management feels that there is scope of improvement in rescue implementation as there is serious risk of asphyxiation due to oxygen deficient environment including presence of toxic and or flammable environment leading to death.



Oxygen Scale

6%	14%	16%	19.5%	21%	More than 23%
Difficult breathing, death in minutes	Faulty judgement, rapid fatigue	Impaired judgement and breathing	Minimum for safe entry	Normal	Oxygen enriched, extreme fire hazard

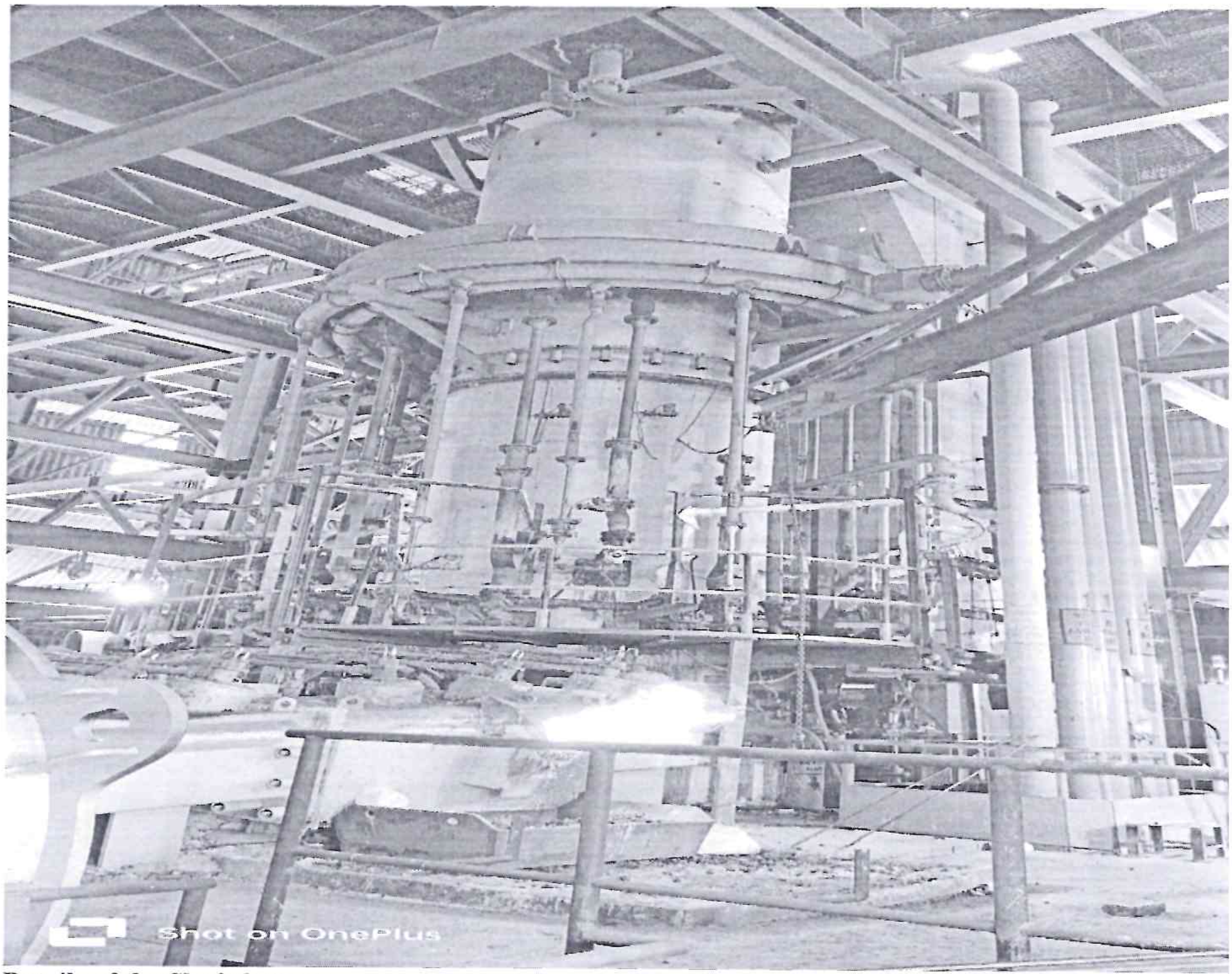


Objective:
To save the person from confined space risk & consequence by ensure timely rescue by-

- Organizing a competent team of rescuer having capability and capacity in various scenarios of rescue
- Arranging adequate rescue equipment and providing hands-on training to rescue team and prepare them for the worst-case scenario
- Organizing 360-degree preparedness encompassing Vedanta team, business partners, Occupational health specialist, ambulance services, tie-up hospital and other support services

The high-risk activities carried out in Confined space (actual photos) where rescue preparedness and mitigation are mandatory are as follows:

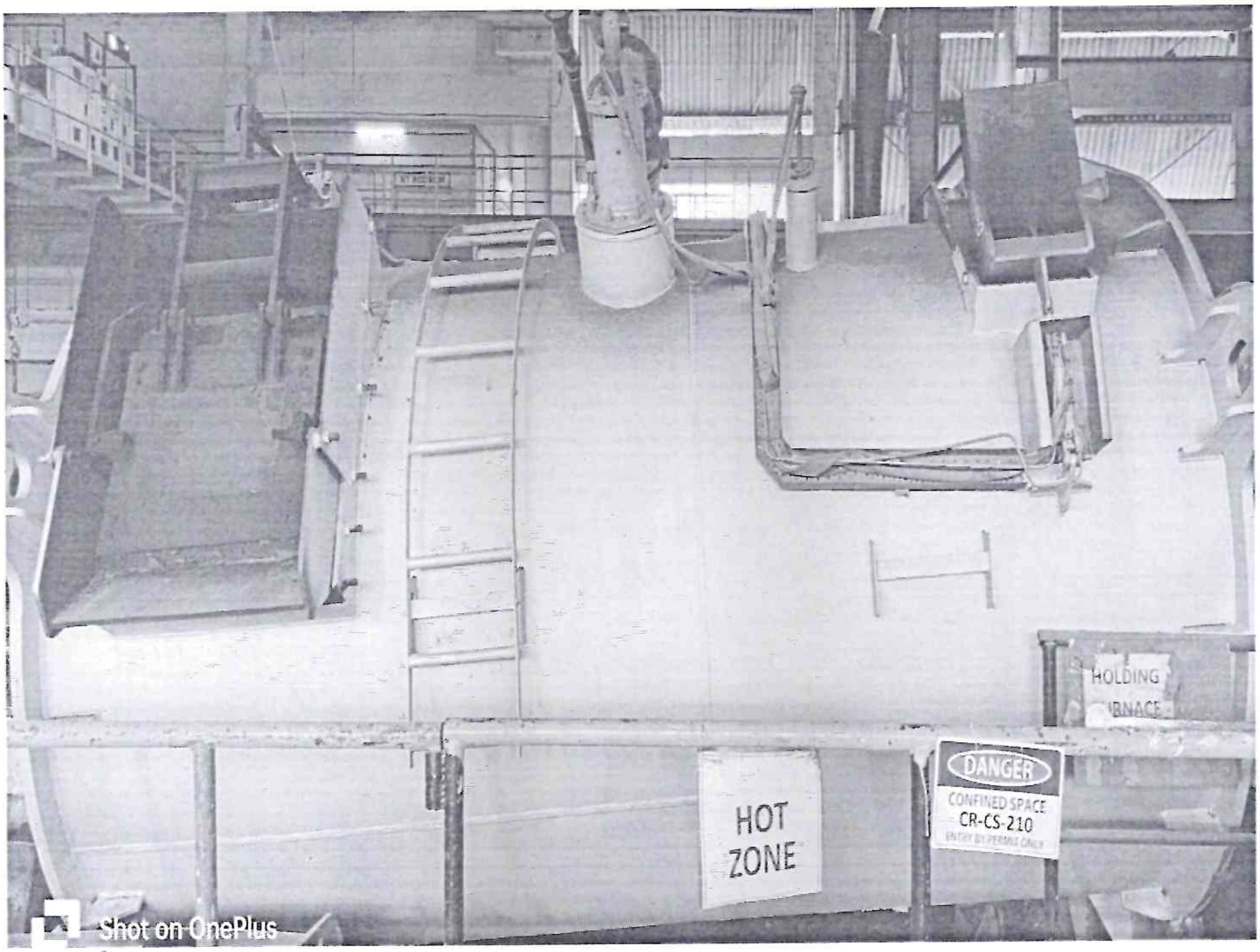
Shaft furnaces:



Details of the Shaft furnace-

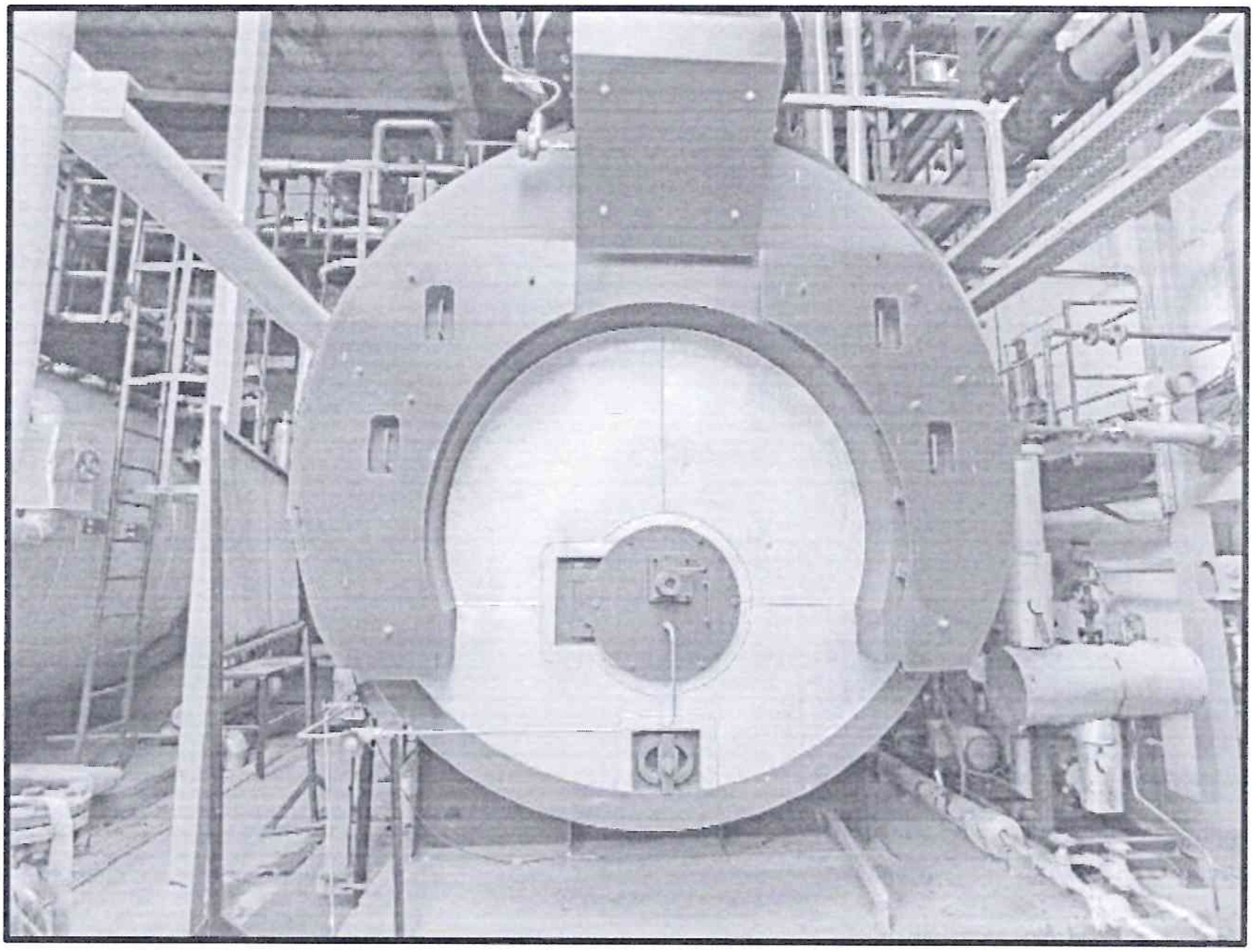
Sr. No.-
Location-
Dimension (in meters)-
Confined space activity-
Process involved-
Hazards-
Frequency of working inside confined space (in days)-
Site In-charge-
How many workers are involved-

Holding Furnace:



Details of the Holding furnace-
Sr. No.-
Location-
Dimension (in meters)-
Confined space activity-
Process involved-
Hazards-
Frequency of working inside confined space (in days)-
Site In-charge-
How many workers are involved

Boilers:



Details of the Boiler-

Sr. No.-

Location-

Dimension (in meters)-

Confined space activity-

Process involved-

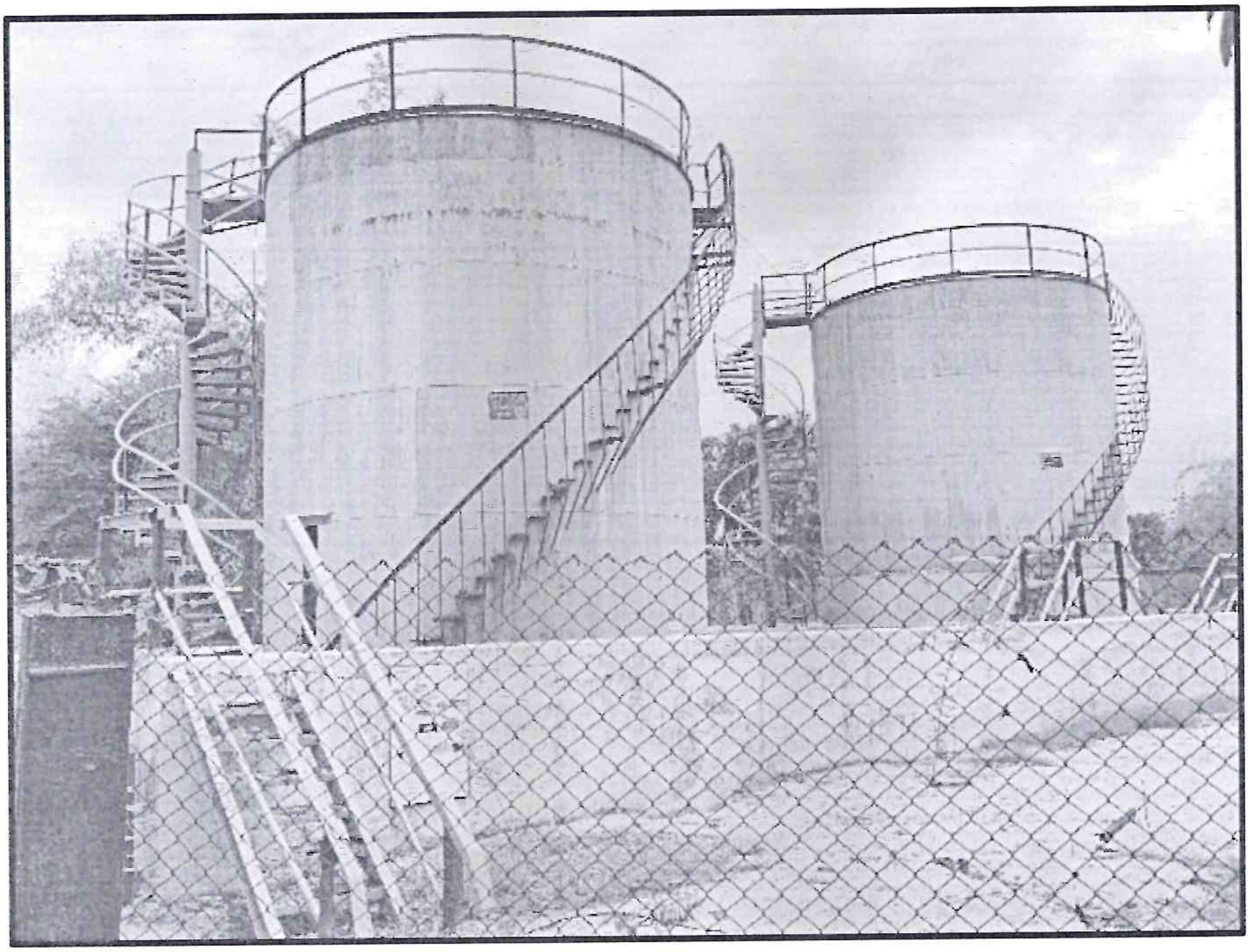
Hazards-

Frequency of working inside confined space (in days)-

Site In-charge-

How many workers are involved-

Furnace oil storage tanks-



Details of the Furnace Oil storage tank-

Sr. No.-

Location-

Dimension (in meters)-

Confined space activity-

Process involved-

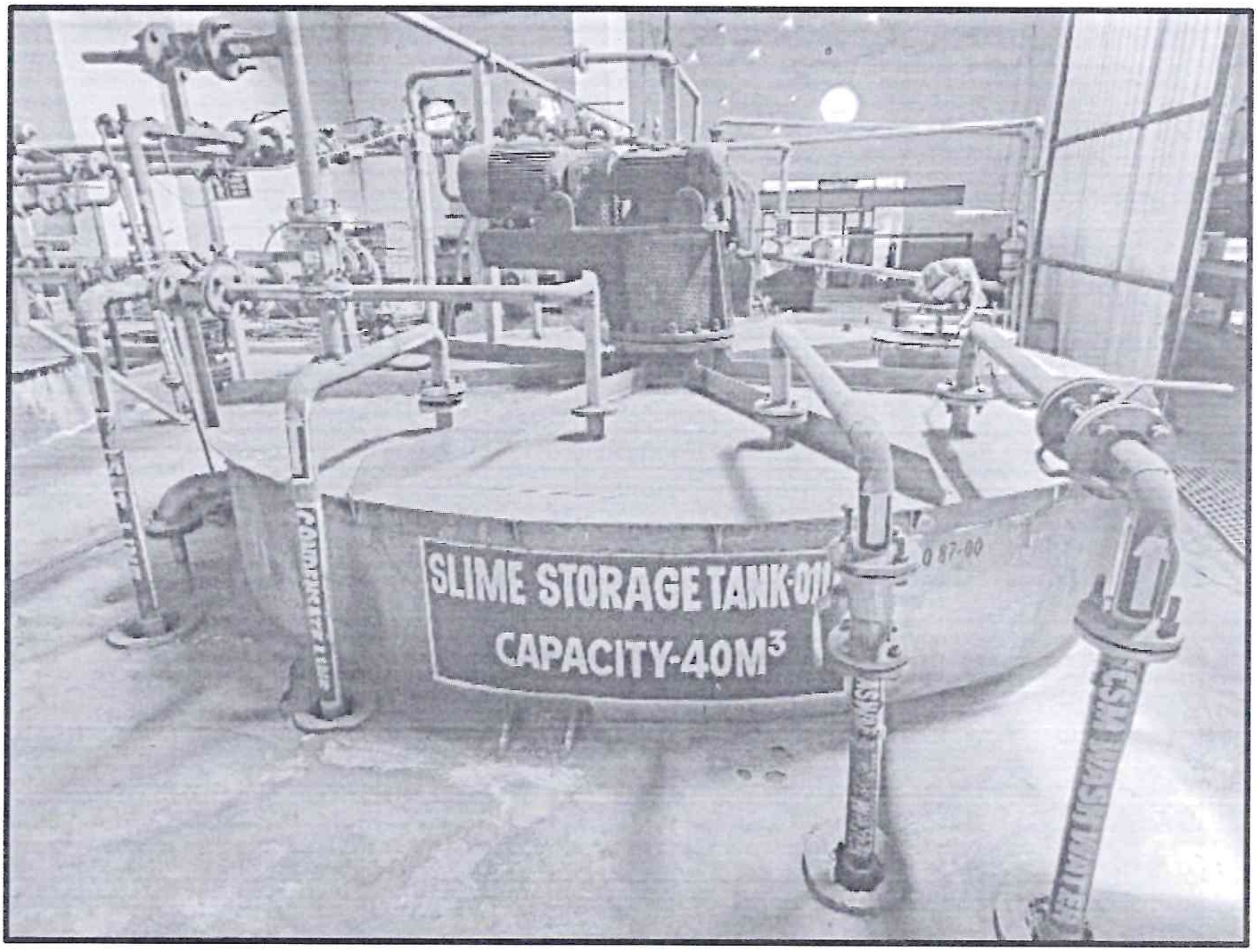
Hazards-

Frequency of working inside confined space (in days)-

Site In-charge-

How many workers are involved-

Slime storage tanks-



Details of the Slime storage tank-

Sr. No.-

Location-

Dimension (in meters)-

Confined space activity-

Process involved-

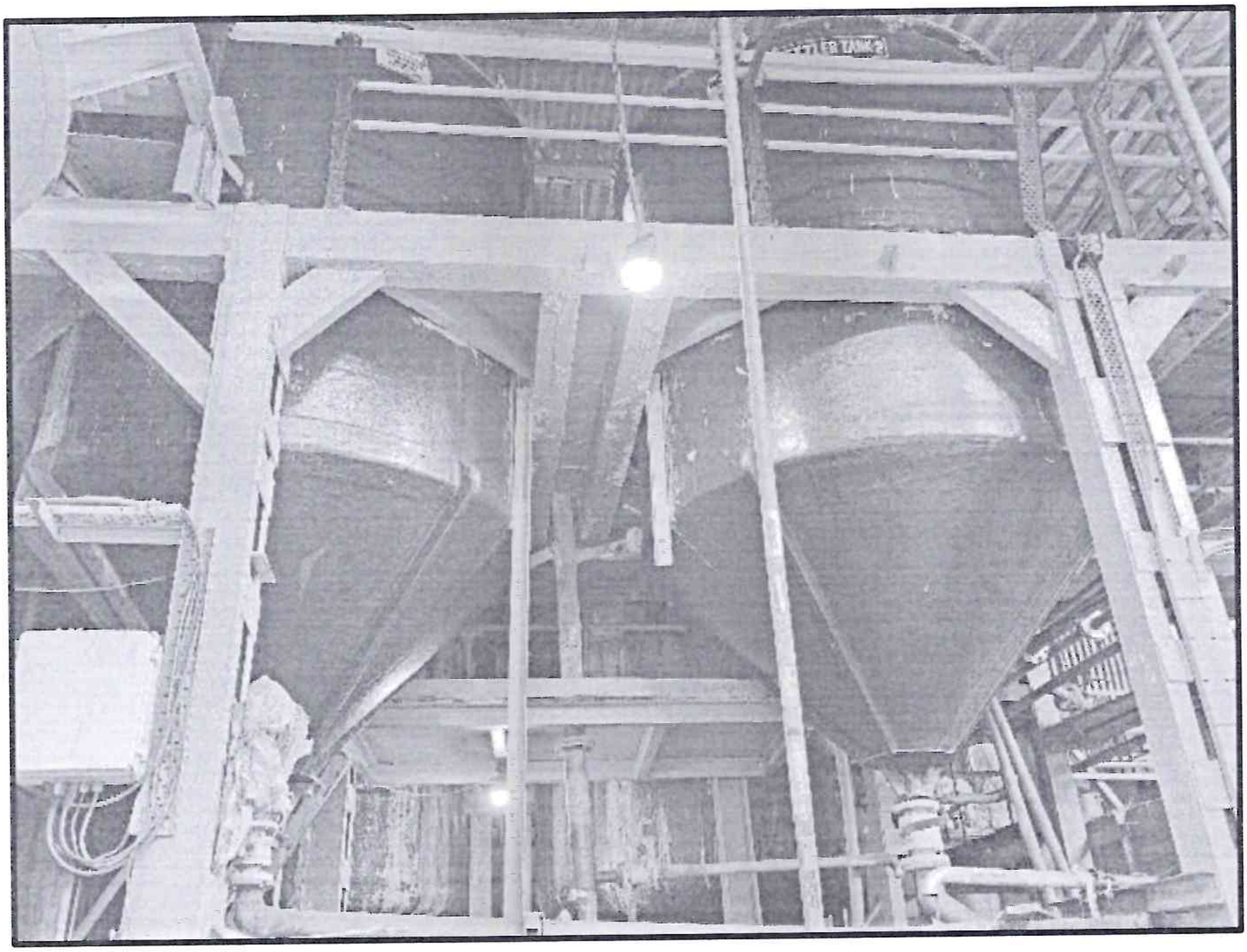
Hazards-

Frequency of working inside confined space (in days)-

Site In-charge-

How many workers are involved-

Settler tank 1 & 2-



Details of the Settler tank-

Sr. No.-

Location-

Dimension (in meters)-

Confined space activity-

Process involved-

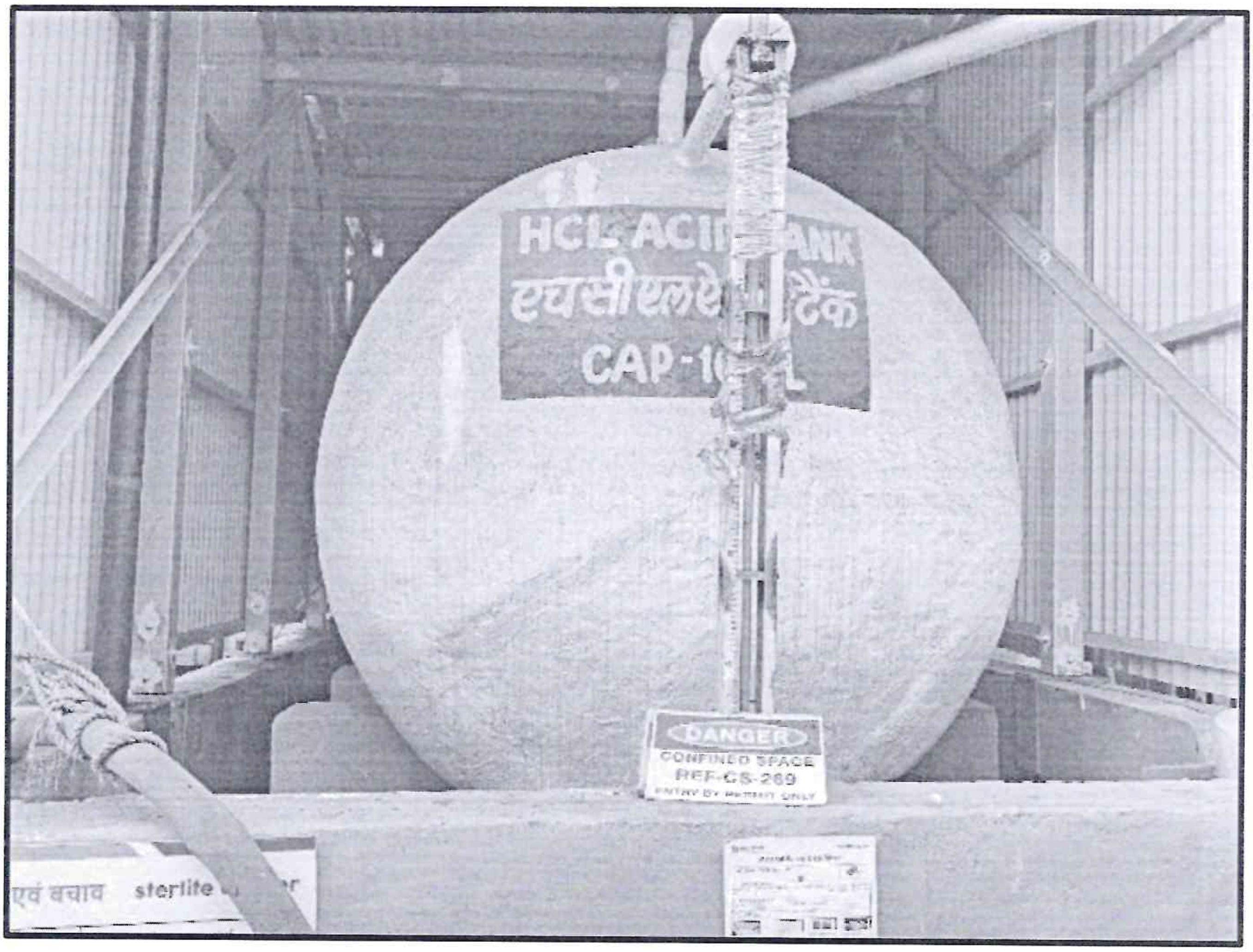
Hazards-

Frequency of working inside confined space (in days)-

Site In-charge-

How many workers are involved-

Hydrochloric acid storage tank-



Details of the Hydrochloric acid tank-

Sr. No.-

Location-

Dimension (in meters)-

Confined space activity-

Process involved-

Hazards-

Frequency of working inside confined space (in days)-

Site In-charge-

How many workers are involved-

Sulphuric acid storage tank-

Details of the Sulphuric acid tank-

Sr. No.-

Location-

Dimension (in meters)-

Confined space activity-

Process involved-

Hazards-

Frequency of working inside confined space (in days)-

Site In-charge-

How many workers are involved-

Sodium Hydroxide storage tank-

Details of the Sodium Hydroxide storage tank-

Sr. No.-

Location-

Dimension (in meters)-

Confined space activity-

Process involved-

Hazards-

Frequency of working inside confined space (in days)-

Site In-charge-

How many workers are involved-

Introduction of Rescue procedure

While operating in confined spaces, it is essential for safety experts to contemplate what to do in case an individual gets hurt, incapacitated, or stuck and needs to be rescued.
More than just helping to understand why this happens, this plan will assist in preventing a fatality caused by asphyxiation and or injured caused by flammable environment.

Here are **seven steps** for confined space rescue procedures that will assist safety experts in planning and executing a rescue from a confined space quickly and safely. Every second counts.

1. Conduct rescue drills

One of the best ways to prepare is to practice. There are many theories about the best practices for confined space rescue and rescue drills allow safety professionals to put those theories to the test.

Rescue drills provide personnel the experience of working through different scenarios in order to familiarize themselves with situations they could encounter in confined spaces. Conducting rescue drills helps prepare teams for working in confined spaces, and when necessary, rescuing coworkers.

2. Understand that rescues fall into two categories

There are time-sensitive and non-time-sensitive rescues. Time-sensitive or "emergency" rescues typically involve oxygen-deficient atmospheres where there is a small window of time, typically 10 minutes, to get someone out.

An example of a non-time-sensitive rescue would be a situation in which someone falls and breaks an ankle going into a confined space. In these types of circumstances there are sufficient O_2 levels and, therefore, the rescue is not as time-sensitive and can be conducted without the use of supplemental oxygen.

Understanding both types of rescues help safety professionals develop strategies for implementing and executing an appropriate response.

3. Make every entrant wear a full body harness

Of all the equipment involved in confined space rescue, perhaps the most important is the full body harness. Many rescues require lifting equipment to remove a person from a confined space, and that lifting equipment will need to attach to a full body harness.

The harness plays an important role in both vertical rescues to help lift a worker out of a space, and horizontal rescues to help place the worker on a stretcher or rescue board.

Without a full body harness, rescues can become much more difficult and time-consuming. Workers do not want to have to place a full body harness on an inert body, particularly if it is a time-sensitive rescue.

4. Survey confined spaces for rescue

Rescuing someone in a timely manner requires an in-depth knowledge of the parameters of the confined space. Depending on the configuration and location of a space, the safety professional may need to adjust the rescue strategy.

For instance, the traditional tripod used to lift workers out may not be feasible in all confined spaces. As such, anchor eyebolts may need to be installed over the confined space entry points for vertical rescues. Surveying beforehand will help determine the proper tools and techniques for working within a confined space and getting someone out in a rescue situation.

5. Survey openings

Along with surveying confined spaces, it's also important to survey and assess openings. Some may think that a rescue person with a self-contained breathing apparatus (SCBA) can fit into any confined space, and/or be able to move freely within it. In many instances, that is not the case.

Surveying openings provides an assessment of how much room workers and/or rescuers will have to enter a confined space, and what types of equipment they will be able to bring with them.

6. Meet with local authorities about rescue capabilities

It is important to remember that it is not always possible to rely on Emergency Response Support System (ERSS)- 112 is a PAN India number for providing emergency support such as Police, fire and ambulance, for a confined space rescue solution. Depending on the situation, authorities may not have the manpower or capability to perform a rescue.

Understanding the capabilities of local authorities helps safety professionals develop rescue plans suited to each particular situation. If local authorities are not able to assist, the safety professional must adjust accordingly to ensure that the proper personnel are in place to respond to an emergency.

7. Have a rescue team

In many rescue situations, workers think that they can hook a person up to the line on the rescue winch and don't need a rescue team.

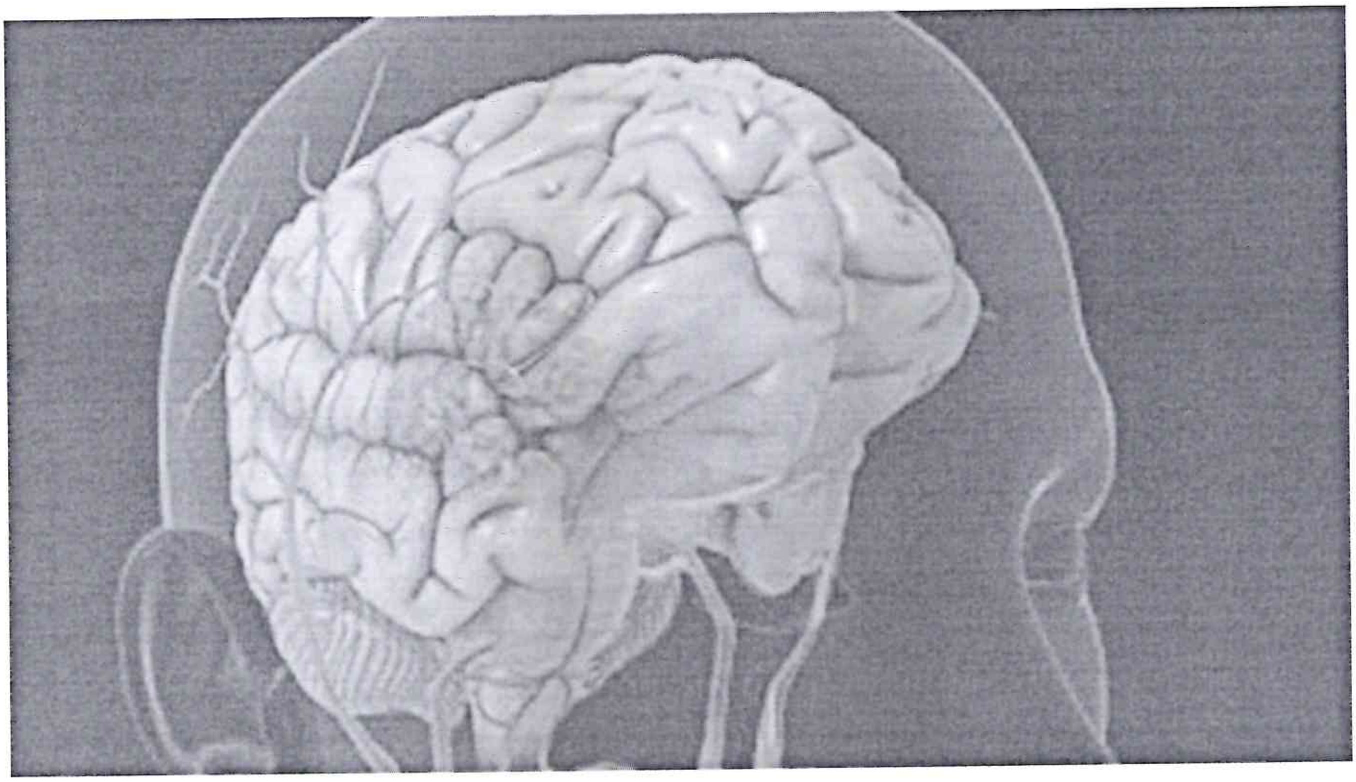
While this strategy might work in situations where someone goes straight to the bottom of a confined space and does not move, those types of incidents are rare. Furthermore, if there is more than one entrant, that strategy is not feasible.

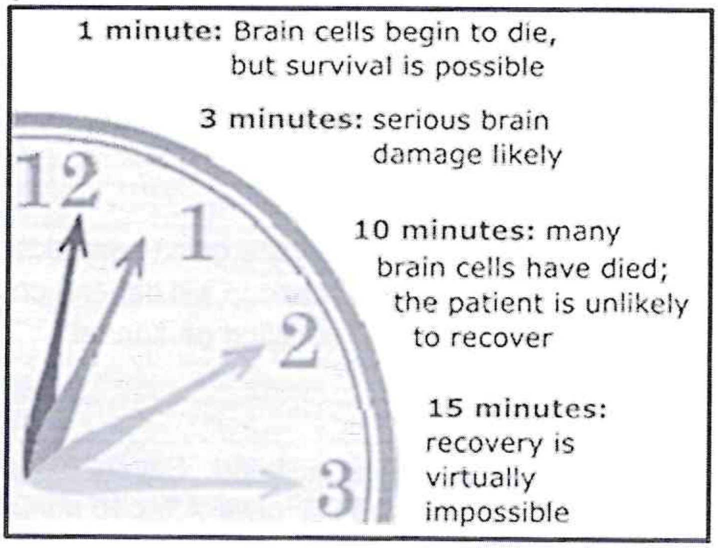
Safety professionals need to ensure that they have fully equipped and trained confined space rescue team (CSRT) ready to respond in an emergency situation. Regardless of the confined space or opening, a rescue team needs to be prepared and ready to respond in a timely manner if someone is injured, trapped or incapacitated.

CSRTs could be qualified members of an employer's own team, local emergency response or an employee of Business partner. The team's makeup will depend on several factors including budget, local resources and the availability of qualified personnel.

The Time Frame

After being unconscious, the employee is incapacitated. Severe oxygen deprivation can cause life-threatening problems including coma and seizures. After 10 minutes without oxygen, brain death occurs. Brain death means there is no brain activity. A person needs life support measures like a mechanical ventilator to help them breathe and stay alive.





The Need for a Rescue Plan

The person faces considerable difficulty in breathing due to oxygen deficiency and exposure to toxic gas and may die eventually due to asphyxiation if the person is not rescued within the first 10 minutes. Also there is chances of fire incident due to hot work like welding/ grinding/ gas-cutting in presence of flammable environment.

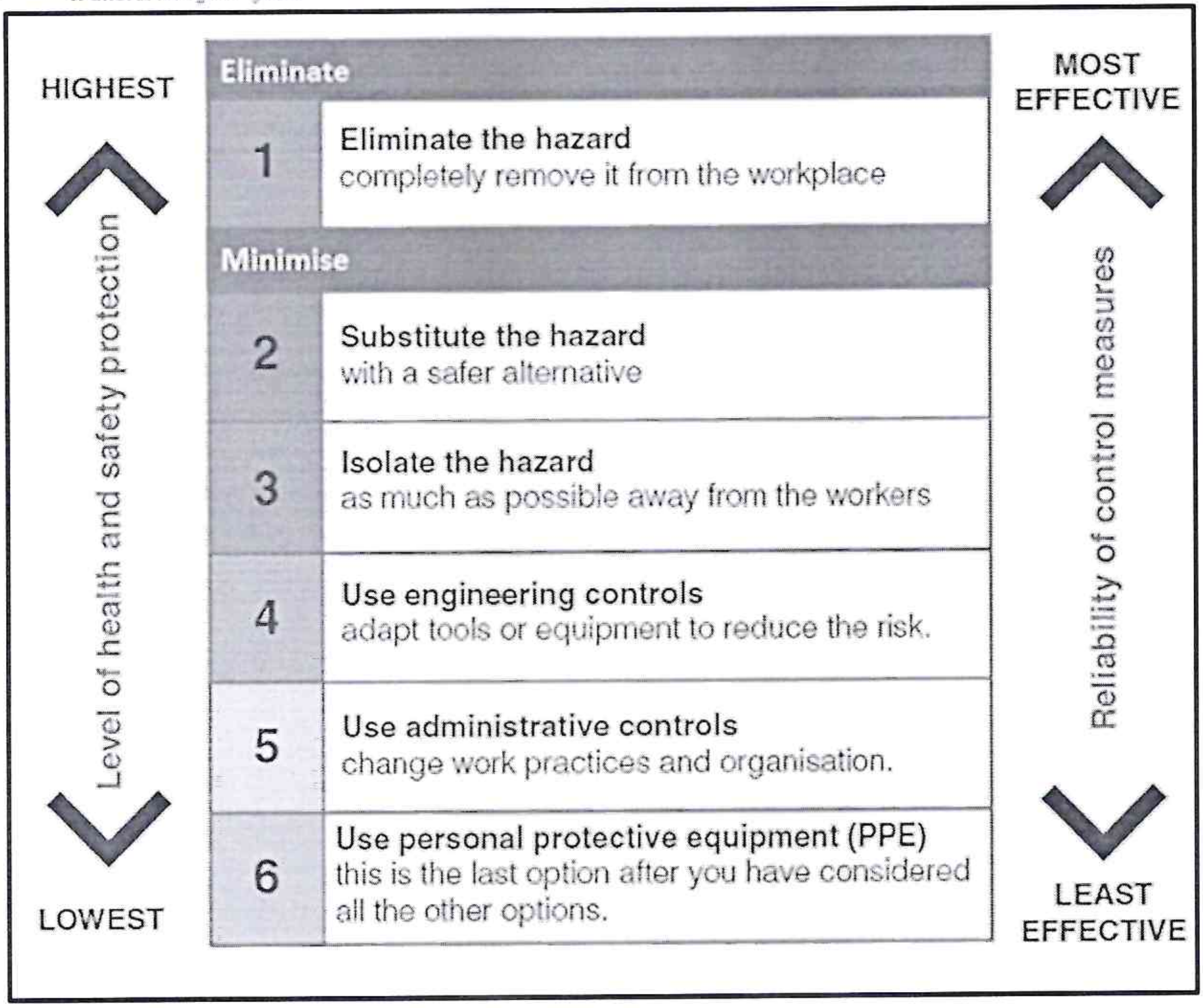
Through a thought-out, detailed and fully implemented rescue plan, we can save the precious life of the person. It is now a legal requirement of the as per Factories Act 1948 to have a rescue plan. The best rescue strategy is to take every possible precaution to prevent such scenarios in the first place.

But, the lack of any form of a pre-conceived confined space rescue plan not only puts the victim at risk but also puts rescuers in harm's way.

Whenever there are unplanned attempts to rescue, second or third injuries or fatalities may not be uncommon.

Develop and Implement Hierarchy of controls based on the risk assessment to reduce the risk level and consequence.

Focus on Elimination and substitution but it is not possible in most of the cases of Confined space so combination of Engineering, administrative and PPE controls shall be implemented.

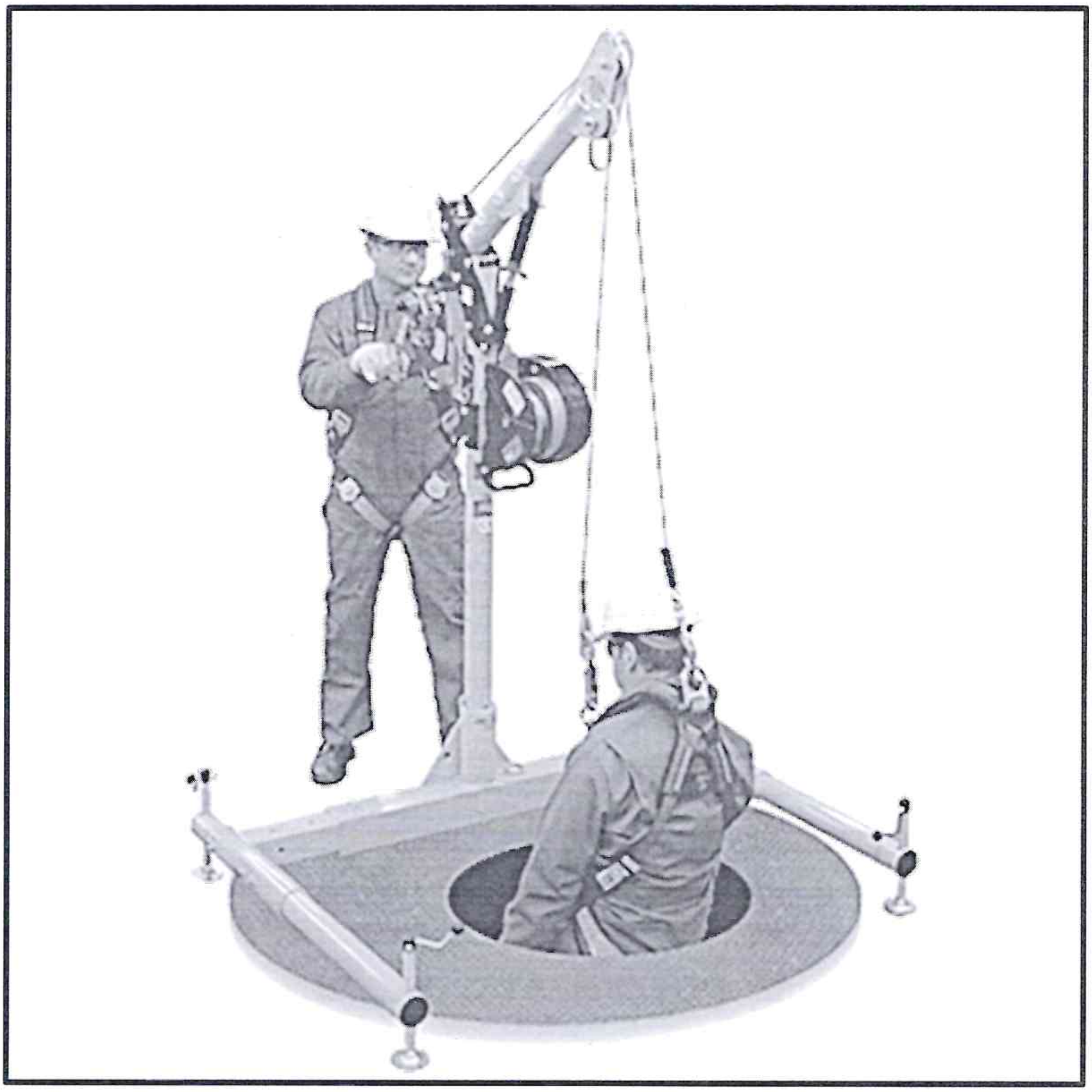


Engineering controls-

Following Engineering controls shall be implemented as mentioned below to reduce the likelihood and probability of the risk.

- Use of multi-gas detectors to monitor atmosphere within the confined space before entry and during confined space work at regular intervals 90 minutes at 3 different levels i.e., 0.5 meters from top level of the vessel, at center of the vessel and 0.5 meters from bottom surface of the vessel
- Vertical vessels- Use of Tripod stand with vertical ascend rescue kit equipped with vertical stretcher, lifeline to descend, rope ladder, motorized winch.
- Horizontal vessels- Use vertical stretcher along with motorized winch if the height of manway opening is 2 meters above the ground level.
- Provide adequate illumination as per the environment within the confined space. Use 24 V hand lamp only as a std. procedure. If the environment is flammable, then use flame proof and explosion proof equipment.

- Provide local exhaust ventilation in case 2 manholes are available. If only one manway is available, then provide forced draught of air for maintaining fresh air and also diluting any toxic or flammable gas and maintaining adequate oxygen levels in the confined space environment.
- In each of the vertical vessels, davit swivel arm shall be fitted retractable motorized winch for fast rescue.

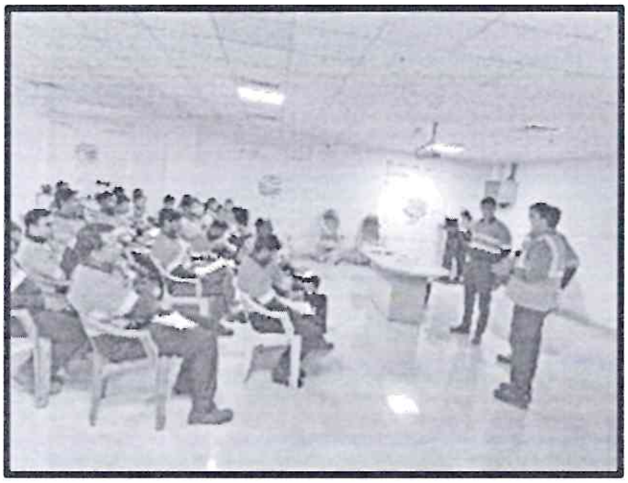


Administrative Control-

- Implement Confined space Procedure and Permit to work procedure for sensitizing all stakeholders and monitoring the confined space activity for ensuring safe entry and exit of all employees/ business partners.
- Provide training about hazards, risks and control measures and emergency rescue procedure of Confined space and check effectiveness post training and while at work.
- Provide authorized gas tester on the job to monitor confined space atmosphere for any presence of flammable gas, toxic gas, and oxygen deficiency before and during the confined space activity.
- Provide an attendant who will coordinate and communicate with the employees and business partners and ensure their safety.
- Walkie talkie sets shall be provided for effective communication with the attendant and rescuers.
- Ambulance equipped with necessary life-saving aids like AED (Automatic External Defibrillator), pulse, Oxygen level, blood pressure monitor and Occupational health specialist shall be kept on stand-by
- Adequate rest intervals shall be provided to all people working inside confined space after 20 minutes of working inside confined space. It shall be the responsibility of the attendant to ensure that adequate rest intervals are provided to the person.
- Competent rescue teams shall be kept on stand-by for mitigating any emergency. Adequate nos. of ABC Fire extinguishers of 6 kg capacity shall be kept ready along with SCBA, ascend rescue kit.

Identify and train at least a team of 6 rescuers from each dept. in different scenarios of Confined space rescue and monitor their effectiveness by measuring the reduction in time required to rescue. Follow the PDCA (Plan- Do- Check- Act) for continual improvement.

- Provide special uniform to nominated rescuers for ease in identification.
- Provide roles and responsibilities to all rescuers so that each rescuer knows their responsibility during the actual rescue process.
- Identify Rescue team leader and Dy. Rescue team leader who will be in-charge of commanding and monitor the effectiveness rescue process.
- Reward and recognize the Rescue team to boost their morale and provide incentives as per organization HR process (for employees) or as per contract (for business partners).

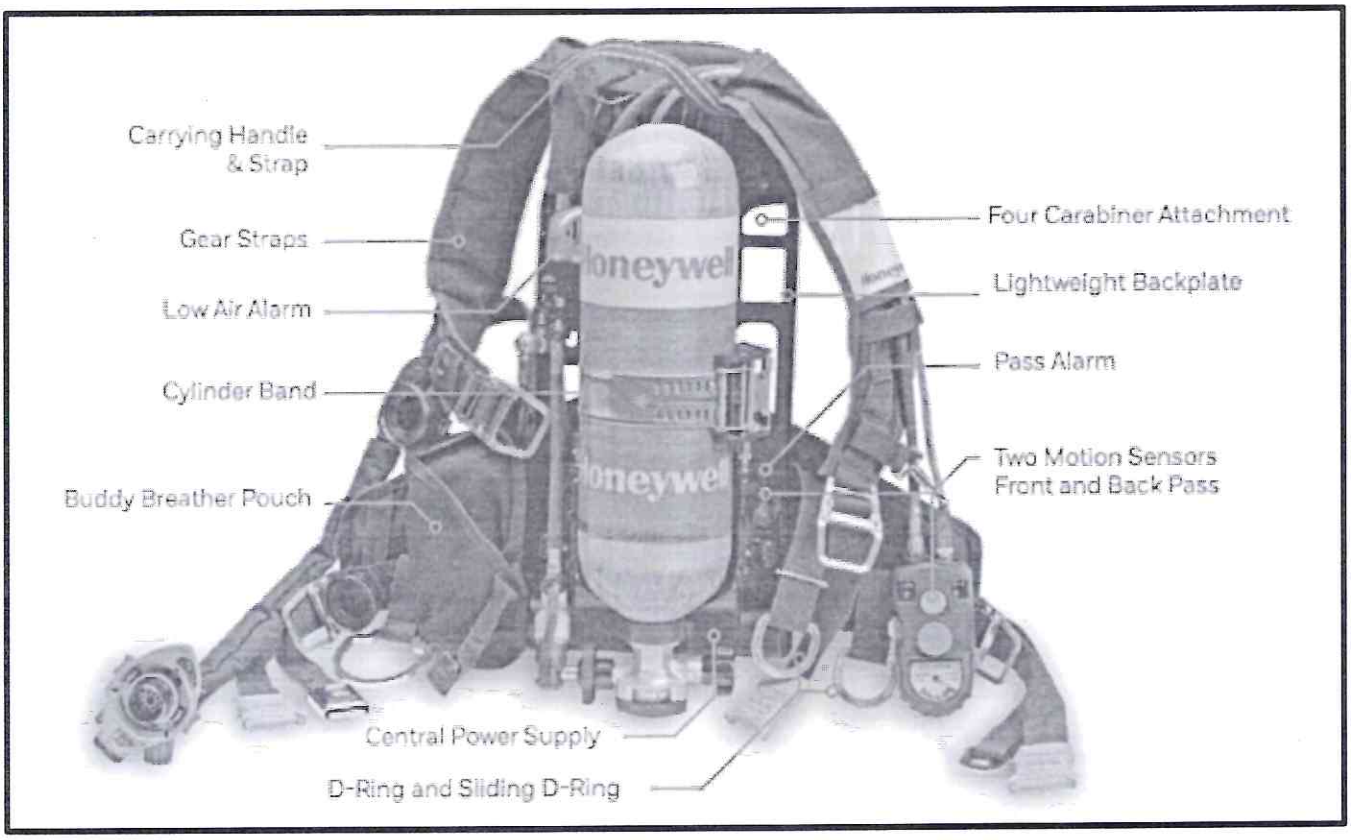


Personal Protective Equipment (PPE)-

Although PPE is the line resort or last line of defense, it is mandatory so ensure that all entrants and rescue team are always wearing the Full body harness with twin lanyard and scaffold snap hook.

Other mandatory PPE required are:

- Safety helmet with chin strap and head mounted lamp.
- Reflective jacket
- Safety shoes
- Safety goggles
- N95 mask
- SCBA if Purging gas like N2 is used for controlling the flammable environment.
- SCBA (Self Contained Breathing Apparatus) is mandatory for rescue team.
- Cut resistant hand gloves.



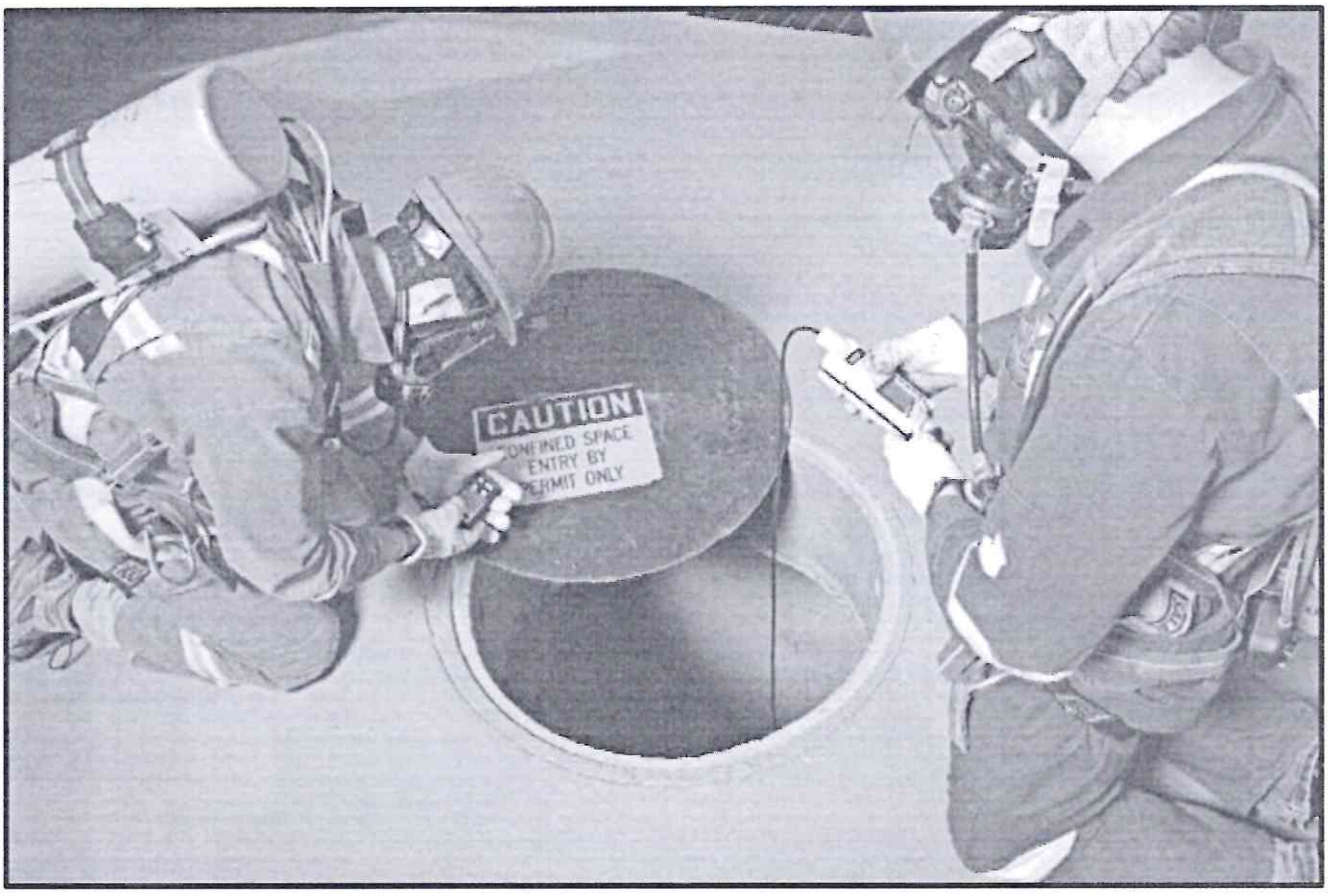
Self-Contained Breathing Apparatus (SCBA)

Critical Phases of Rescue

Listed below are the five critical phases of rescuing a suspended employee:

1) Before entering the confined space- Preventive stage
2) During working inside confined space
3) Asphyxiation or Incapacitated
4) Rescue
5) Post Rescue

Each rescue phase presents unique safety challenges. Whatever training employees have received will determine how they respond to different phases.



Before entering Confined space

- **REVIEW:** Carefully review the space to be entered, types of hazards, safety precautions, control measures and assess the work to be done in the space.
- **COMPLETE:** Obtain and fully complete a Permit-Required Confined Space Entry form.
- **ENSURE:** Ensure all participants in the entry have been trained. Assign duties to all participants.
- **IDENTIFY:** Proper PPE, ventilation, air monitoring and rescue equipment needed for entry is available, assembled, and in working order.
- **COMMUNICATE:** Determine method of communication between entrants and attendants.
- **VERIFY:** Air monitors have been calibrated and are in working order.

- **ALERT:** Call the emergency rescue service to alert them of the planned confined space entry.
- **ELIMINATE:** Eliminate any condition making it unsafe to remove the entrance cover.
- **GUARD:** After opening confined space, immediately guard the area so no one can fall in when open. Erect signs, barriers, and barrier tape as necessary. Provide traffic control if necessary.
- **LOCK-OUT:** Before entering, lock and tag out (LOTO) any mechanical hazards, gas, water lines, or electrical power. Try to turn the system on after locking it out to ensure that there is no energy available.
- **CHECK:** Set up and check out all rescue equipment (e.g., tripod, body harness, rescue line) prior to entry.
- **MONITOR THE AIR:** Monitor the air in space to ensure that no atmospheric hazards are present. Monitoring must always be conducted immediately before entering any confined space.

Safe Air Limits for Confined Space Entry:

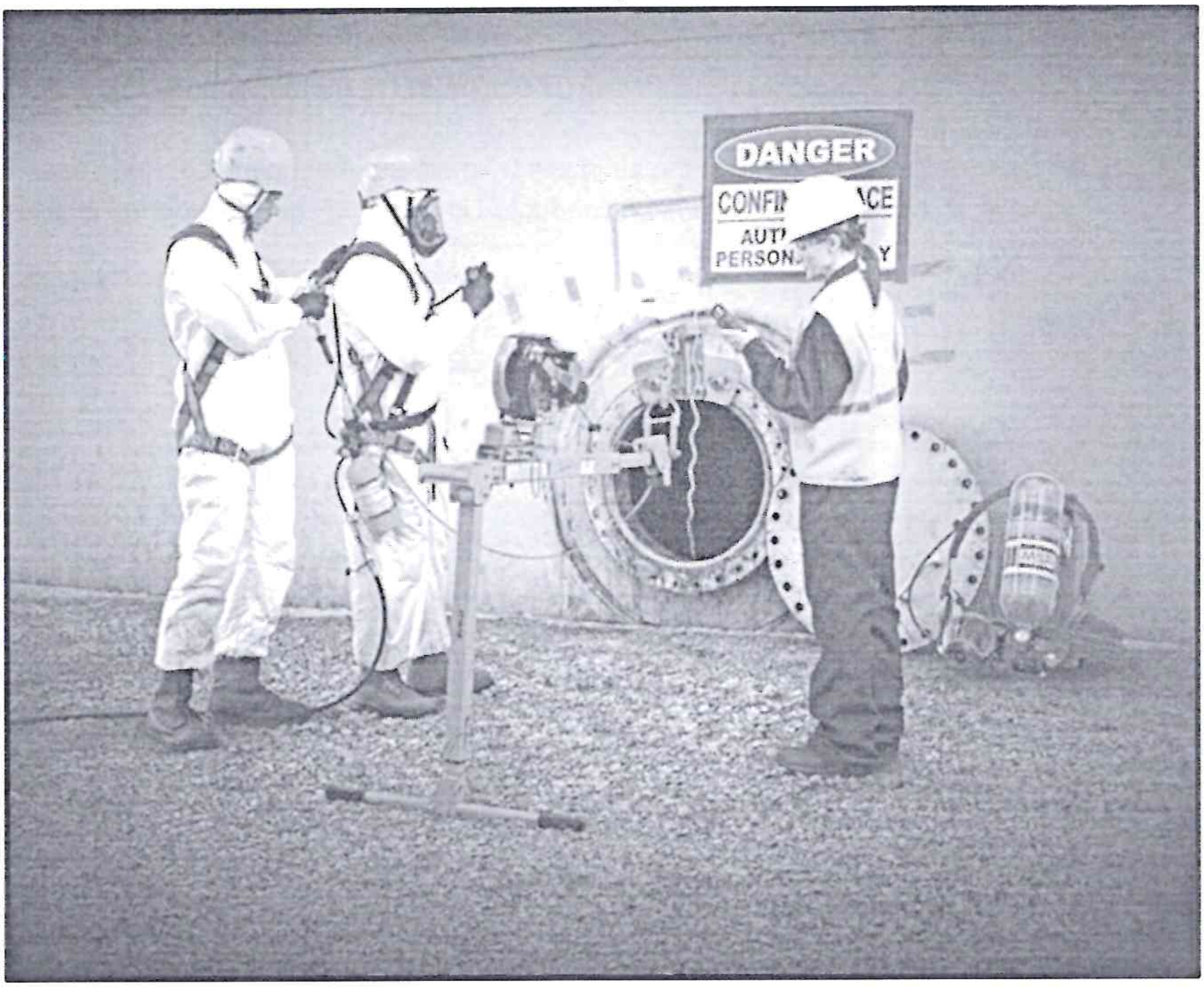
- Oxygen: Between 19.5% and 23.5%
- LEL: Less than 10% of known LEL (LEL = Lower Explosive Limit)
- H2S: Less than 10 ppm
- CO: Less than 30 ppm

If the air quality inside the confined space is safe with no fresh air ventilation, and if the only hazard is an actual or potentially hazardous atmosphere that can be controlled by continuous forced air ventilation, the space may be reclassified temporarily as an "Alternate Entry Procedures" space.

- **RECORD:** Write air monitoring results and time taken on the Permit.
- **VENTILATE:** If air monitoring results or the work to be done in space indicates that ventilation is needed, set up the ventilation blower so that it takes in clean, uncontaminated air. Do not place intake next to vehicles, gas-powered equipment, or exhaust vent from a lab or other potentially contaminated work area. Attach ducting to blower, turn on, and place exhaust end well inside confined space. Run the blower for

the minimum amount of time needed to purge the space based on the calculations made on the Permit.

- **RETEST:** After ventilation, check the atmosphere in the space again to confirm that acceptable entry conditions are present.
- **SIGN:** If all entry conditions are met, the entry supervisor signs the permit allowing entry to the space.
- **CONNECT:** Entrant puts on Full body harness and connects to lifeline from tripod and secures communication device if necessary.



During working inside Confined space

- **ENTER:** Entrant enters the space and checks for hazards that may not have been detected.
- **MONITOR:** Attendant continues to monitor the atmosphere in the space throughout entry and record results at least every 90 minutes.
- **COMMUNICATE:** Attendant communicates with entrant for entire period while the entrant is working inside confined space
- **EMERGENCY EXIT:** Exit the space immediately if any of the following occurs:
  - A hazardous atmosphere is detected.
  - Any health or safety hazard is detected.
  - If the entrant shows signs of exposure to atmospheric hazards, feels ill, notices strong odors or has other safety concerns.
  - Re-evaluate space and/ or modify entry procedure before re-entering.
- **NORMAL EXIT:** When work is completed, return space to proper condition and secure opening.
- **FILE:** Return the Entry Permit to the entry supervisor for filing.

Asphyxiation

Asphyxiation or suffocation occurs when the body is deprived of Oxygen. There are different types of Asphyxia.

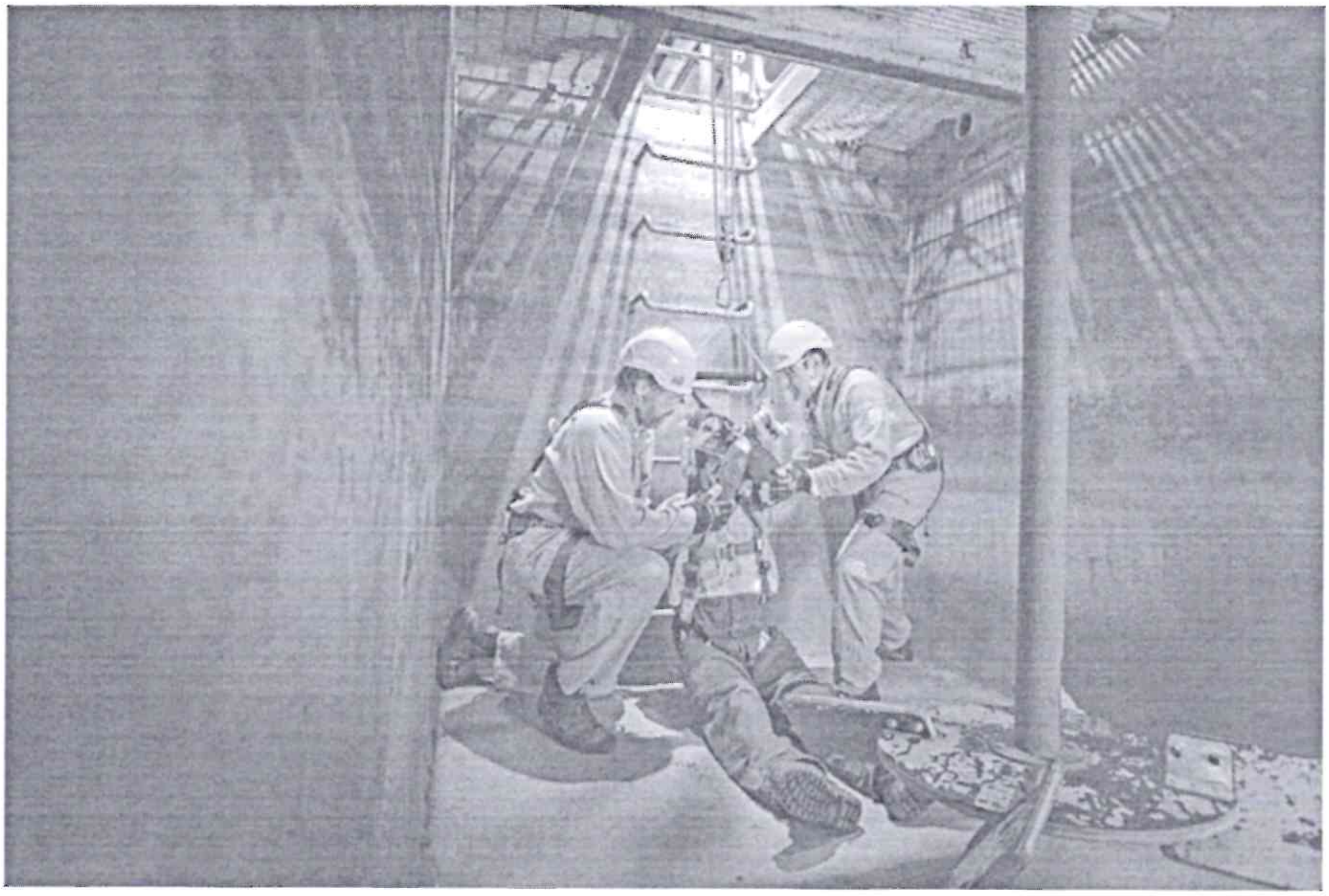
O2 Asphyxiation- It can happen due to lack of Oxygen.

Chemical asphyxia- It can happen due generation of CO within the confined space.

Rescue

Rescue must come rapidly to minimize the dangers of Asphyxiation. Rescuers must always be ready for any eventuality.

- First check the atmosphere with a gas tester and ensure SCBA with respirator or respirator with direct supply of Oxygen.
- There is no point in risking the life of rescuers so the Rescue team lead will monitor the operation continuously and communicate effectively with the rescuers.
- Use Motorized winch or Tripod ascender system to rescue the incapacitated person.



Post Rescue

Treatment for asphyxiation depends on the cause. It may include:

Cardiopulmonary resuscitation (CPR)- CPR is a procedure that involves chest compressions to promote blood and oxygen circulation. It's used when a person's heart stops beating.

Oxygen therapy- Oxygen therapy delivers oxygen to the lungs of an incapacitated person. It may involve a ventilator, a breathing tube, or a mask or nose tube that provides oxygen.

Medication- Rush the person to tie-up hospital and coordinate with specialist for advice and continue with CPR.

Methods of Rescue:

The different types of rescues that can be adopted based on the scenario are:

1. **Self-Rescue:**
   This procedure is adopted when the entrant feels dizzy while working inside the confined space and he communicates with the attendant about his ill health and then the attendant advises him/ her to immediately exit the confined space.
   Also, there is a chance of fire incident due to carrying out hot works and the entrant extinguishes the fire and exits out of the confined space environment.
   In this case, external assistance is not required by the entrant as he/ she is still capable and gathers energy to exit out of confined space all by himself/ herself.

2. Self-assisted Rescue:

This procedure is adopted when the entrant feels dizziness but is still conscious and seeks help from attendant to support him/ her in rescue process.
Also, there is a chance of fire incident due to carrying out hot works and the entrant requires support to extinguish the fire and exits out of the confined space environment.

In this case the rescue team lowers the rope ladder and uses an ascender system like powered winch and helps the entrant to exit out of the confined space chamber.

3. **Rescue with Ascend rescue kit:**

This procedure is adopted when the entrant is incapacitated and does not respond to the call or communication of the attendant. The attendant informs the rescue team leader to take over the charge of rescue swiftly. Here time is the essence.

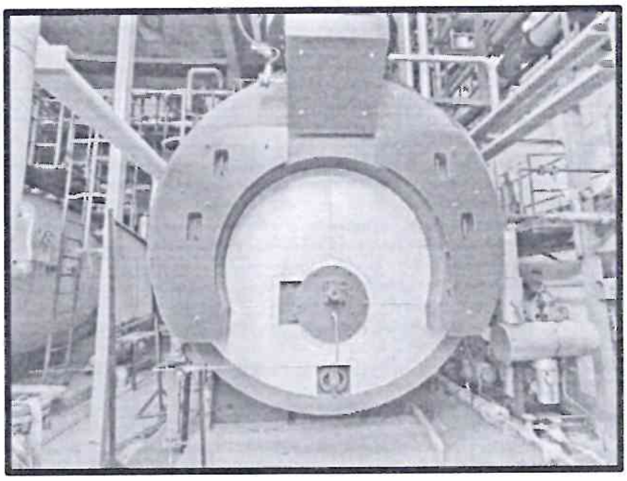
In this procedure, the following steps are involved:

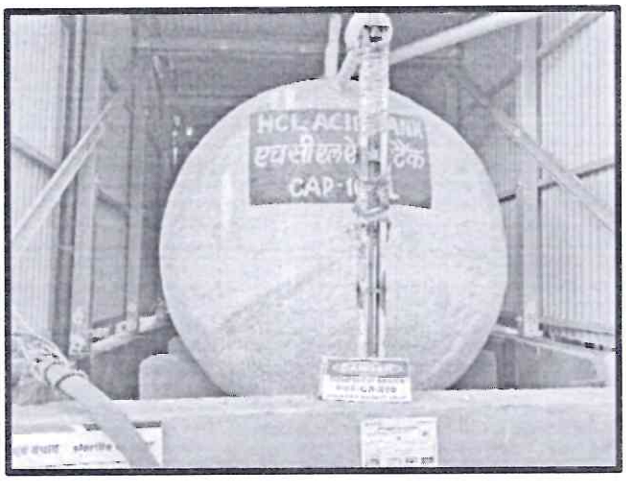
- The rescue team consisting of 2-3 members which is already in stand-by received instruction from their team leader to enter the confined space by wearing SCBA with full respirator.
- Two rescue team members then secures himself by connecting the retractor system and the ascend system (motorized winch)
- The rescuers then swiftly glide down and reach down to the unconscious person along with a vertical stretcher.
- The rescuer then slides the unconscious person on the vertical stretcher secures him firmly with strap and then connects the vertical stretcher with motorized winch.
- The rescuer informs the team leader to ascend the victim.
- The rescue team member at the top lifts the victim and the two members hold the stretcher with guide rope for safe egress of the victim out of the confined space.
- The Rescue team then moves the victim to an Ambulance and starts measuring his pulse and provides CPR and provides O2 supply with manual Oxygen ventilator/ pump and if required moves him to nearest tie-up hospital.

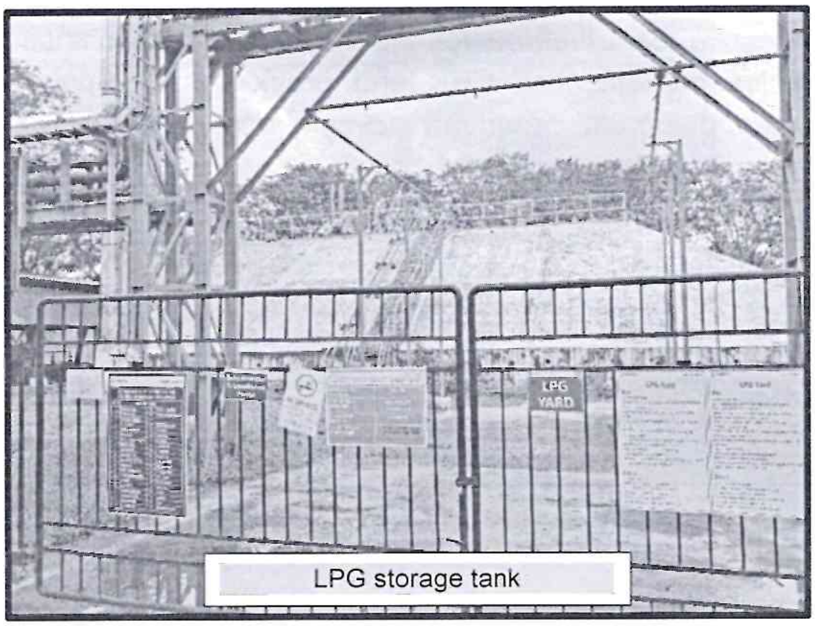
Based on the above rescue methods, we have categorized the confined space into 3 types:

- Horizontal tank confined space
- Vertical tank confined space
- Shaft furnace confined space

Horizontal tank confined space rescue plan (HTCSRP)-







LPG storage tank

The common activities that will be conducted in the horizontal tank during shutdown are as follows:

- Cleaning
- Degreasing
- N2 Purging
- Modification/ maintenance
- Hot work- Welding/ Grinding/ gas-cutting

In all above activities, Risk of asphyxiation due to Oxygen deficiency and presence of traces of Inert gas and or fire incident due to presence of flammable gas in the environment.

Following risk control measures shall be implemented as pre-requisites before starting work to ensure safety of the entrants:

Engineering Controls-

- Use calibrated multi-gas detectors to monitor atmosphere within the confined space before entry and during confined space work at regular intervals 90 minutes at 3 different levels i.e., 0.5 meters from the opening manway of the vessel, at center of the vessel and 0.5 meters from the rear end of the vessel.
- Horizontal vessels- Keep vertical stretcher along with motorized winch on stand-by if the height of manway opening is 2 meters above the ground level.
- Adequate illumination arrangement shall be provided as per the environment within the confined space. Use 24 V hand lamp only as a std. procedure. If the environment is flammable, then use flame proof and explosion proof equipment.
- Local exhaust ventilation shall be provided in case 2 manholes are available. If only one manway is available, then provide forced draught of air for maintaining fresh air and diluting any toxic or flammable gas and maintaining adequate oxygen levels in the confined space environment.

Administrative controls-

- Follow Confined Space Procedure and Permit to work procedure for sensitizing all stakeholders and monitoring the confined space activity for ensuring safe entry and exit of all employee's/ business partners.
- Only competent workers are allowed to work in Confined space and check effectiveness while at work.
- An authorized gas tester shall be on the job to monitor confined space atmosphere before and during the confined space activity.
- Attendant will coordinate and communicate with the employees and business partners and ensure their safety.
- The attendant and entrant shall use Walkie talkie sets for effective communication with the attendant and rescuers.
- Ambulance equipped with necessary life-saving aids like AED (Automatic External Defibrillator), pulse, Oxygen level, blood pressure monitor and Occupational health specialist shall be kept on stand-by
- Attendant shall ensure that adequate rest intervals shall be provided to all persons working inside confined space after 20 minutes of working inside confined space.
- The entrant shall keep a 6 kg. ABC type fire extinguisher and a fire blanket if hot work is carried out to extinguish any fire and mitigate the emergency.
- Rescue team comprising of 2 competent rescuers shall be kept on stand-by who have been adequately trained to handle horizontal tank confined space rescue.

Personal Protective Equipment (PPE)-

Although PPE is the line resort or last line of defense, it is mandatory so ensure that all entrants and rescue team is always wearing the Full body harness with twin lanyard and scaffold snap hook.

Other mandatory PPE required are:

- Safety helmet with chin strap and head mounted lamp.
- Reflective jacket
- Safety shoes
- Safety goggles
- SCBA if Purging gas like N2 is used for controlling the flammable environment.
- SCBA (Self Contained Breathing Apparatus) is mandatory for rescue team.
- Cut resistant hand gloves.

Even after above precautions, there is negligible scope for any untoward incident but as we say

"Prevention is better than cure."

In case of eventuality,

Note: Entrant means competent person who is allowed entry to work in confined space and has the required permit to work

Scenario-1

Fire incident during hot work and the person is engulfed in fire.

Rescue plan is as follows:

- The first step will be the entrant will roll over the floor of the tank/ boiler or wrap himself with the fire blanket to extinguish the fire.
- The one of the rescue team members will swiftly enter the tank with additional ABC type of fire extinguisher and will use it to extinguish any remains of the fire.
- Then the rescue team member will safely escort the entrant out of the confined space
- If in case the entrant is injured due to fire and cannot move, then the attendant provides the vertical stretcher to the rescue team members who are inside the confined space, and he will then secure the injured person (Entrant) on the vertical stretcher.
- The Rescue team leader will intimate the Ambulance driver, paramedic team and HSE team leader about the scenario who will reach the place of incident within 3 minutes.
- The HSE team leader will intimate the CEO, Plant Director and site in-charge and brief them about the scenario while on the move to the location of incident.
- The attendant along with other co-workers will pull the stretcher and the rescue team members will support the stretcher from another end and bring the injured person outside of the confined space.
- The Occupational health specialist will place him in ambulance and will check the injured person and provide ointment and clean his wounds while on the move to the nearest tie-up hospital. The Ambulance is equipped with AED, pulse, blood pressure monitoring, SPo2 monitoring instrumentation, O2 manual pump and O2 cylinder.

- The HSE team leader will call the administration dept of tie-up hospital who in turn will keep the necessary aids ready for treatment of injured person.
- Once the injured person reaches the tie-up hospital, the Occupational health specialist will coordinate with resident doctor and if required call specialists for trauma management.
- The Site in-charge in coordination with HR will nominate an administration personnel to ensure that the best treatment is provided to the injured person till he reports back to work.
- The HR personnel will then intimate the family members of the injured person and provide information about the incident and explain how management is taking adequate care for speedy recovery.
- The confined space will be secured by the security guard and entry point to the tank shall be sealed.
- The accident investigation team shall conduct a detailed investigation identifying the root causes of the incident.
- The HSE team leader shall adequately reward and recognize the Rescue team members for motivation and empowerment.

Scenario-2
Asphyxiation of entrant due to Oxygen deficiency and or presence of toxic gas

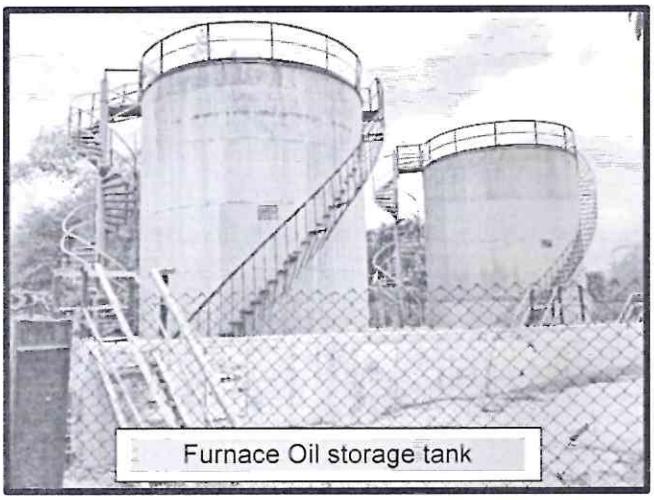
Rescue plan is as follows:

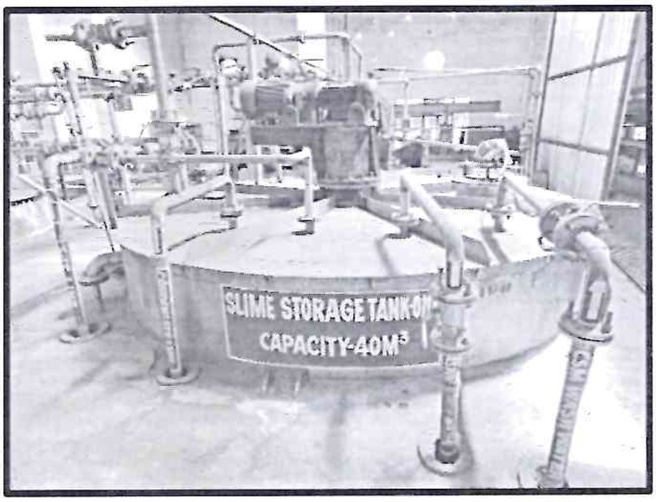
- First check the confined space atmosphere with a gas tester and ensure Self Contained Breathing Apparatus (SCBA) with respirator or respirator with direct supply of Oxygen.
- There is no point in risking the life of rescuers so the Rescue team lead will monitor the operation continuously and communicate effectively with the rescuers.
- If in case the entrant is unconscious due to lack of O2 and cannot move, then the attendant provides the vertical stretcher to the rescue team members who are inside the confined space, and he will then secure the injured person (Entrant) on the vertical stretcher.
- The Rescue team leader will intimate the Ambulance driver, paramedic team and HSE team leader about the scenario who will reach the place of incident within 3 minutes.
- The HSE team leader will intimate the CEO, Plant Director and site in-charge and brief them about the scenario while on the move to the location of incident.
- The attendant along with other co-workers will pull the stretcher and the rescue team members will support the stretcher from another end and bring the Unconscious person outside of the confined space.
- The HSE team leader will call the administration dept of tie-up hospital who in turn will keep the necessary aids ready for treatment of unconscious person.
- The Occupational health specialist will place him in ambulance and will check the injured person and provide O2 supply while on the move to the nearest tie-up hospital. The Ambulance is equipped with AED, pulse, blood pressure monitoring, SPo2 monitoring instrumentation, O2 manual pump and O2 cylinder. Continuous monitoring of Spo2 level

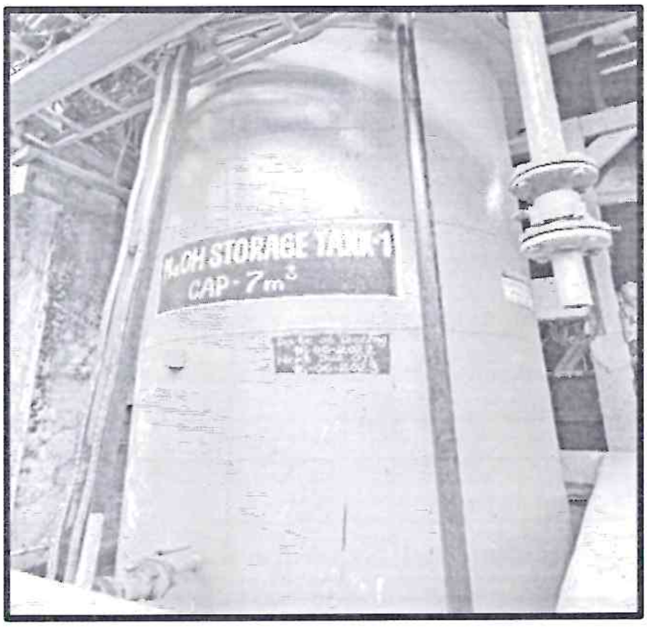
shall be checked by the Occupational Health specialist and update his condition to HSE team leader and to tie-up hospital.

- Occupational Health Specialist shall provide CPR (Cardiopulmonary Resuscitation) to the unconscious person if pulse stops, and heart function stops. Also calibrated shock through AED (Automatic external Defibrillator) shall be given to the unconscious person to revive the heart function.
- Once the injured person reaches the tie-up hospital, the Occupational health specialist will coordinate with resident doctor and if required call specialists for trauma management.
- The Site in-charge in coordination with HR will nominate an administration personnel to ensure that the best treatment is provided to the injured person till he reports back to work.
- The HR personnel will then intimate the family members of the injured person and provide information about the incident and explain how management is taking adequate care for speedy recovery.
- The confined space will be secured by the security guard and entry point to the tank shall be sealed.
- The accident investigation team shall conduct a detailed investigation identifying the root causes of the incident.
- The HSE team leader shall adequately reward and recognize the Rescue team members for motivation and empowerment.

Vertical Tank confined space rescue plan (VTCSRP)


Furnace Oil storage tank





The common activities that will be conducted in the horizontal tank during shutdown are as follows:

- Cleaning
- Degreasing
- N2 Purging
- Modification/ maintenance
- Hot work- Welding/ Grinding/ gas-cutting

In all above activities, Risk of asphyxiation due to Oxygen deficiency and presence of traces of Inert gas and or fire incident due to presence of flammable gas in the environment.

Following risk control measures shall be implemented as pre-requisites before starting work to ensure safety of the entrants:

Engineering Controls-

- Use calibrated multi-gas detectors to monitor atmosphere within the confined space before entry and during confined space work at regular intervals 90 minutes at 3 different levels i.e., 0.5 meters from the opening manway of the vessel, at center of the vessel and 0.5 meters from the rear end of the vessel.
- Vertical vessels- Keep vertical stretcher along with motorized winch on stand-by for rescue and lower the person from height to ground level.
- Keep Tripod Ascend rescue kit with winch/ portable flexible davit arm and extra karabiner sling for vertical stretcher ready over the manhole/ entry point.
- Keep flexible nylon ladder of height 15 meter ready on stand-by
- Adequate illumination arrangement shall be provided as per the environment within the confined space. Use 24 V hand lamp only as a std. procedure. If the environment is flammable, then use flame proof and explosion proof equipment.
- Local exhaust ventilation shall be provided in case 2 manholes are available. If only one manway is available, then provide forced draught of air for maintaining fresh air and diluting any toxic or flammable gas and maintaining adequate oxygen levels in the confined space environment.

Administrative controls-

- Follow Confined space Procedure and Permit to work procedure for sensitizing all stakeholders and monitoring the confined space activity for ensuring safe entry and exit of all employees/ business partners.
- Only competent workers are allowed to work in Confined space and check effectiveness while at work.
- An authorized gas tester shall be on the job to monitor confined space atmosphere before and during the confined space activity.
- Attendant will coordinate and communicate with the employees and business partners and ensure their safety.
- The attendant and entrant shall use Walkie talkie sets or other way of effective communication with the attendant and rescuers.
- Ambulance equipped with necessary life-saving aids like AED (Automatic External Defibrillator), pulse, Oxygen level, blood pressure monitor and Occupational health specialist shall be kept on stand-by
- Attendant shall ensure that adequate rest intervals shall be provided to all persons working inside confined space after 20 minutes of working inside confined space.
- The entrant shall keep a 6 kg. ABC type fire extinguisher and a fire blanket if hot work is carried out to extinguish any fire and mitigate the emergency.

- Rescue team comprising of 2 competent rescuers shall be kept on stand-by who have been adequately trained to handle horizontal tank confined space rescue.

Personal Protective Equipment (PPE)-
Although PPE is the line resort or last line of defense, it is mandatory so ensure that all entrants and rescue team are always wearing the Full body harness with twin lanyard and scaffold snap hook.
Other mandatory PPE required are:

- Safety helmet with chin strap and head mounted lamp.
- Reflective jacket
- Safety shoes
- Safety goggles
- SCBA if Purging gas like N2 is used for controlling the flammable environment.
- SCBA (Self Contained Breathing Apparatus) is mandatory for rescue teams.
- Cut resistant hand gloves.

Even after above precautions, there is negligible scope for any untoward incident but as we say, **"Prevention is better than cure."**

In case of eventuality,

Note: Entrant means competent person who is allowed entry to work in confined space and has the required permit to work

Scenario-1
Fire incident during hot work and the person is engulfed in fire.

Rescue plan is as follows:

- The first step will be the entrant will roll over the floor of the tank/ boiler or wrap himself with the fire blanket to extinguish the fire.
- The one of the rescue team members will swiftly enter the tank with the help of flexible nylon ladder and the other person will lower the additional ABC type of fire extinguisher which will be use it to extinguish any remains of the fire.
- Then the rescue team member will safely escort the entrant out of the confined space
- If in case the entrant is injured due to fire and cannot move then the attendant lowers the vertical stretcher to the rescue team members with the help of winch wire rope who are inside the confined space and the rescuer will then secure the injured person (Entrant) on the vertical stretcher and strap him firmly and attach the karabiner sling on head side of the vertical stretcher and a guide rope at feet side of the stretcher
- The rescuer will attach the karabiner sling to the winch and then signal the attendant for lifting the vertical stretcher by means of winch secured to the tripod or powered winch anchored to the portable flexible davit arm as shown on page no. 19.

- The Rescue team leader will inform the Ambulance driver, paramedic team and HSE team leader about the scenario who will reach the place of incident within 3 minutes.
- The HSE team leader will intimate the CEO, Plant Director and site in-charge and brief them about the scenario while on the move to the location of incident.
- The attendant along with other co-workers will pull the stretcher and the rescue team members will support the stretcher from another end and bring the injured person outside of the confined space.
- The portable flexible davit arm will be then used to lower the vertical stretcher from the top of the tank with the help of winch.
- Once the stretcher reaches the ground, the rescue team and co-workers will untie the stretcher and place the injured person on the Ambulance stretcher.
- The Occupational health specialist will place him in ambulance and will check the injured person and provide ointment and clean his wounds while on the move to the nearest tie-up hospital. The Ambulance is equipped with AED, pulse, blood pressure monitoring, SPo_2 monitoring instrumentation, O_2 manual pump and O_2 cylinder.
- The HSE team leader will call the administration dept of tie-up hospital who in turn will keep the necessary aids ready for treatment of injured person.
- Once the injured person reaches the tie-up hospital, the Occupational health specialist will coordinate with resident doctor and if required call specialists for trauma management.
- The Site in-charge in coordination with HR will nominate an administration personnel to ensure that the best treatment is provided to the injured person till he reports back to work.
- The HR personnel will then intimate the family members of the injured person and provide information about the incident and explain how management is taking adequate care for speedy recovery.
- The confined space will be secured by the security guard and entry point to the tank shall be sealed.
- The accident investigation team shall conduct a detailed investigation identifying the root causes of the incident.
- The HSE team leader shall adequately reward and recognize the Rescue team members for motivation and empowerment.

Scenario-2
Asphyxiation of entrant due to Oxygen deficiency and or presence of toxic gas

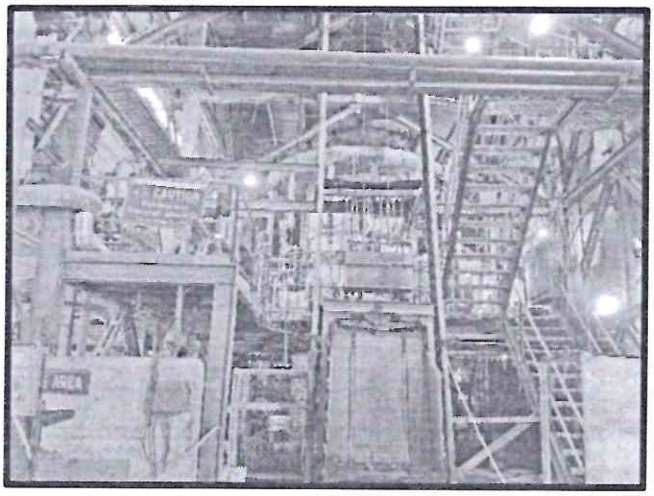
Rescue plan is as follows:

- First check the confined space atmosphere with a gas tester and ensure Self Contained Breathing Apparatus (SCBA) with respirator or respirator with direct supply of Oxygen.
- There is no point in risking the life of rescuers so the Rescue team lead will monitor the operation continuously and communicate effectively with the rescuers.
- If in case the entrant is unconscious due to lack of O_2 and is incapacitated, then the attendant lowers the vertical stretcher to the rescue team members with the help of winch

secured on Ascend Tripod rescue kit who are inside the confined space, and he will then secure the injured person (Entrant) on the vertical stretcher.

- The Rescue team leader will inform the Ambulance driver, paramedic team and HSE team leader about the scenario who will reach the place of incident within 3 minutes.
- The HSE team leader will intimate the CEO, Plant Director and site in-charge and brief them about the scenario while on the move to the location of incident.
- The attendant along with other co-workers will pull the stretcher with the help of winch and the rescue team members will guide the vertical stretcher from other end by guide rope and bring the Unconscious person outside of the confined space.
- The portable flexible davit arm will be then used to lower the vertical stretcher from the top of the tank with the help of winch.
- Once the stretcher reaches the ground, the rescue team and co-workers will untie the stretcher and place the injured person on the Ambulance stretcher.
- The HSE team leader will call the administration dept of tie-up hospital who in turn will keep the necessary aids ready for treatment of unconscious person.
- The Occupational health specialist will place him in ambulance and will check the injured person and provide O_2 supply while on the move to the nearest tie-up hospital. The Ambulance is equipped with AED, pulse, blood pressure monitoring, SPo2 monitoring instrumentation, O_2 manual pump and O_2 cylinder. Continuous monitoring of Spo2 level shall be checked by the Occupational Health specialist and update his condition to HSE team leader and to tie-up hospital.
- Occupational Health Specialist shall provide CPR (Cardiopulmonary Resuscitation) to the unconscious person if pulse stops, and heart function stops. Also calibrated shock through AED (Automatic external Defibrillator) shall be given to the unconscious person to revive the heart function.
- Once the injured person reaches the tie-up hospital, the Occupational health specialist will coordinate with resident doctor and if required call specialists for trauma management.
- The Site in-charge in coordination with HR will nominate an administration personnel to ensure that the best treatment is provided to the injured person till he reports back to work.
- The HR personnel will then intimate the family members of the injured person and provide information about the incident and explain how management is taking adequate care for speedy recovery.
- The confined space will be secured by the security guard and entry point to the tank shall be sealed.
- The accident investigation team shall conduct a detailed investigation identifying the root causes of the incident.
- The HSE team leader shall adequately reward and recognize the Rescue team members for motivation and empowerment.

Shaft Furnace Confined Space Rescue Plan-



The activities carried out are as follows:

- Refractory brick lining replacement
- Cleaning and maintenance
- Hot work (Welding/ Grinding/ Gas-cutting)

In all above activities, Risk of asphyxiation due to Oxygen deficiency and presence of traces of Inert gas and or fire incident due to presence of flammable gas in the environment.

Following risk control measures shall be implemented as pre-requisites before starting work to ensure safety of the entrants:

Engineering Controls-

- Use calibrated multi-gas detectors to monitor atmosphere within the confined space before entry and during confined space work at regular intervals 90 minutes at 3 different levels i.e., 0.5 meters from the opening manway of the vessel, at center of the vessel and 0.5 meters from the rear end of the vessel.
- Vertical vessels- Keep vertical stretcher along with motorized winch on stand-by for rescue and lower the person from height to ground level.
- Keep Tripod Ascend rescue kit with winch/ powered motorized winch anchored to fixed structure and extra karabiner sling for vertical stretcher ready over the manhole/ entry point.
- Keep flexible nylon ladder of height 15 meter ready on stand-by
- Adequate illumination arrangement shall be provided as per the environment within the confined space. Use 24 V hand lamp only as a std. procedure. If the environment is flammable, then use flame proof and explosion proof equipment.
- Local exhaust ventilation shall be provided in case 2 manholes are available. If only one manway is available, then provide forced draught of air for maintaining fresh air and diluting any toxic or flammable gas and maintaining adequate oxygen levels in the confined space environment.

Administrative controls-

- Follow Confined Space Procedure and Permit to work procedure for sensitizing all stakeholders and monitoring the confined space activity for ensuring safe entry and exit of all employee's/ business partners.
- Only competent workers are allowed to work in Confined space and check effectiveness while at work.
- An authorized gas tester shall be on the job to monitor confined space atmosphere before and during the confined space activity.
- Attendant will coordinate and communicate with the employees and business partners and ensure their safety.
- The attendant and entrant shall use Walkie talkie sets for effective communication with the attendant and rescuers.
- Ambulance equipped with necessary life-saving aids like AED (Automatic External Defibrillator), pulse, Oxygen level, blood pressure monitor, and Occupational health specialist shall be kept on stand-by
- Attendant shall ensure that adequate rest intervals shall be provided to all persons working inside confined space after 20 minutes of working inside confined space.
- The entrant shall keep a 6 kg. ABC type fire extinguisher and a fire blanket if hot work is carried out to extinguish any fire and mitigate the emergency.
- Rescue team comprising of 2 competent rescuers shall be kept on stand-by who have been adequately trained to handle horizontal tank confined space rescue.

Personal Protective Equipment (PPE)-

Although PPE is the line resort or last line of defense, it is mandatory so ensure that all entrants and rescue team is always wearing the Full body harness with twin lanyard and scaffold snap hook.

Other mandatory PPE required are:

- Safety helmet with chin strap and head mounted lamp.
- Reflective jacket
- Safety shoes
- Safety goggles
- SCBA if Purging gas like N2 is used for controlling the flammable environment.
- SCBA (Self Contained Breathing Apparatus) is mandatory for rescue teams.
- Cut resistant hand gloves.

Even after above precautions, there is negligible scope for any untoward incident but as we say, **"Prevention is better than cure."**

In case of eventuality,

Note: Entrant means competent person who is allowed entry to work in confined space and has the required permit to work

Scenario-1

Fire incident during hot work and the person is engulfed in fire.

Rescue plan is as follows:

- The first step will be the entrant will roll over the floor of the tank/ boiler or wrap himself with the fire blanket to extinguish the fire.
- The one of the rescue team members will swiftly enter the tank with the help of flexible nylon ladder and the other person will lower the additional ABC type of fire extinguisher which will be use it to extinguish any remains of the fire.
- Then the rescue team member will safely escort the entrant out of the confined space
- If in case the entrant is injured due to fire and cannot move then the attendant lowers the vertical stretcher to the rescue team members with the help of winch wire rope who are inside the confined space and the rescuer will then secure the injured person (Entrant) on the vertical stretcher and strap him firmly and attach the karabiner sling on head side of the vertical stretcher and a guide rope at feet side of the stretcher
- The rescuer will attach the karabiner sling to the winch and then signal the attendant for lifting up the vertical stretcher by means of winch secured to the tripod or powered winch anchored to the portable flexible davit arm as shown on page no. 19.
- The Rescue team leader will inform the Ambulance driver, paramedic team and HSE team leader about the scenario who will reach the place of incident within 3 minutes.
- The HSE team leader will intimate the CEO, Plant Director and site in-charge and brief them about the scenario while on the move to the location of incident.
- The attendant along with other co-workers will pull the stretcher and the rescue team members will support the stretcher from another end and bring the injured person outside of the confined space.
- The portable flexible davit arm will be then used to lower the vertical stretcher from the top of the tank with the help of winch.
- Once the stretcher reaches the ground, the rescue team and co-workers will untie the stretcher and place the injured person on the Ambulance stretcher.
- The Occupational health specialist will place him in ambulance and will check the injured person and provide ointment and clean his wounds while on the move to the nearest tie-up hospital. The Ambulance is equipped with AED, pulse, blood pressure monitoring, SPo2 monitoring instrumentation, O2 manual pump and O2 cylinder.
- The HSE team leader will call the administration dept of tie-up hospital who in turn will keep the necessary aids ready for treatment of injured person.
- Once the injured person reaches the tie-up hospital, the Occupational health specialist will coordinate with resident doctor and if required call specialists for trauma management.

- The Site in-charge in coordination with HR will nominate an administration personnel to ensure that the best treatment is provided to the injured person till he reports back to work.
- The HR personnel will then intimate the family members of the injured person and provide information about the incident and explain how management is taking adequate care for speedy recovery.
- The confined space will be secured by the security guard and entry point to the tank shall be sealed.
- The accident investigation team shall conduct a detailed investigation identifying the root causes of the incident.
- The HSE team leader shall adequately reward and recognize the Rescue team members for motivation and empowerment.

Scenario-2
Asphyxiation of entrant due to Oxygen deficiency and or presence of toxic gas

Rescue plan is as follows:

- First check the confined space atmosphere with a gas tester and ensure Self Contained Breathing Apparatus (SCBA) with respirator or respirator with direct supply of Oxygen.
- There is no point in risking the life of rescuers so the Rescue team lead will monitor the operation continuously and communicate effectively with the rescuers.
- If in case the entrant is unconscious due to lack of O2 and is incapacitated, then the attendant lowers the vertical stretcher to the rescue team members with the help of winch secured on Ascend Tripod rescue kit who are inside the confined space, and he will then secure the injured person (Entrant) on the vertical stretcher.
- The Rescue team leader will inform the Ambulance driver, paramedic team and HSE team leader about the scenario who will reach the place of incident within 3 minutes.
- The HSE team leader will intimate the CEO, Plant Director and site in-charge and brief them about the scenario while on the move to the location of incident.
- The attendant along with other co-workers will pull the stretcher with the help of winch and the rescue team members will guide the vertical stretcher from other end by guide rope and bring the Unconscious person outside of the confined space.
- The portable flexible davit arm will be then used to lower the vertical stretcher from the top of the tank with the help of winch.
- Once the stretcher reaches the ground, the rescue team and co-workers will untie the stretcher and place the injured person on the Ambulance stretcher.
- The HSE team leader will call the administration dept of tie-up hospital who in turn will keep the necessary aids ready for treatment of unconscious person.
- The Occupational health specialist will place him in ambulance and will check the injured person and provide O2 supply while on the move to the nearest tie-up hospital. The Ambulance is equipped with AED, pulse, blood pressure monitoring, SPo2 monitoring instrumentation, O2 manual pump and O2 cylinder. Continuous monitoring of Spo2 level

shall be checked by the Occupational Health specialist and update his condition to HSE team leader and to tie-up hospital.

- Occupational Health Specialist shall provide CPR (Cardiopulmonary Resuscitation) to the unconscious person if pulse stops, and heart function stops. Also calibrated shock through AED (Automatic external Defibrillator) shall be given to the unconscious person to revive the heart function.
- Once the injured person reaches the tie-up hospital, the Occupational health specialist will coordinate with resident doctor and if required call specialists for trauma management.
- The Site in-charge in coordination with HR will nominate an administration personnel to ensure that the best treatment is provided to the injured person till he reports back to work.
- The HR personnel will then intimate the family members of the injured person and provide information about the incident and explain how management is taking adequate care for speedy recovery.
- The confined space will be secured by the security guard and entry point to the tank shall be sealed.
- The accident investigation team shall conduct a detailed investigation identifying the root causes of the incident.
- The HSE team leader shall adequately reward and recognize the Rescue team members for motivation and empowerment.

First Aid Procedures (Common)

The key recommendations are:

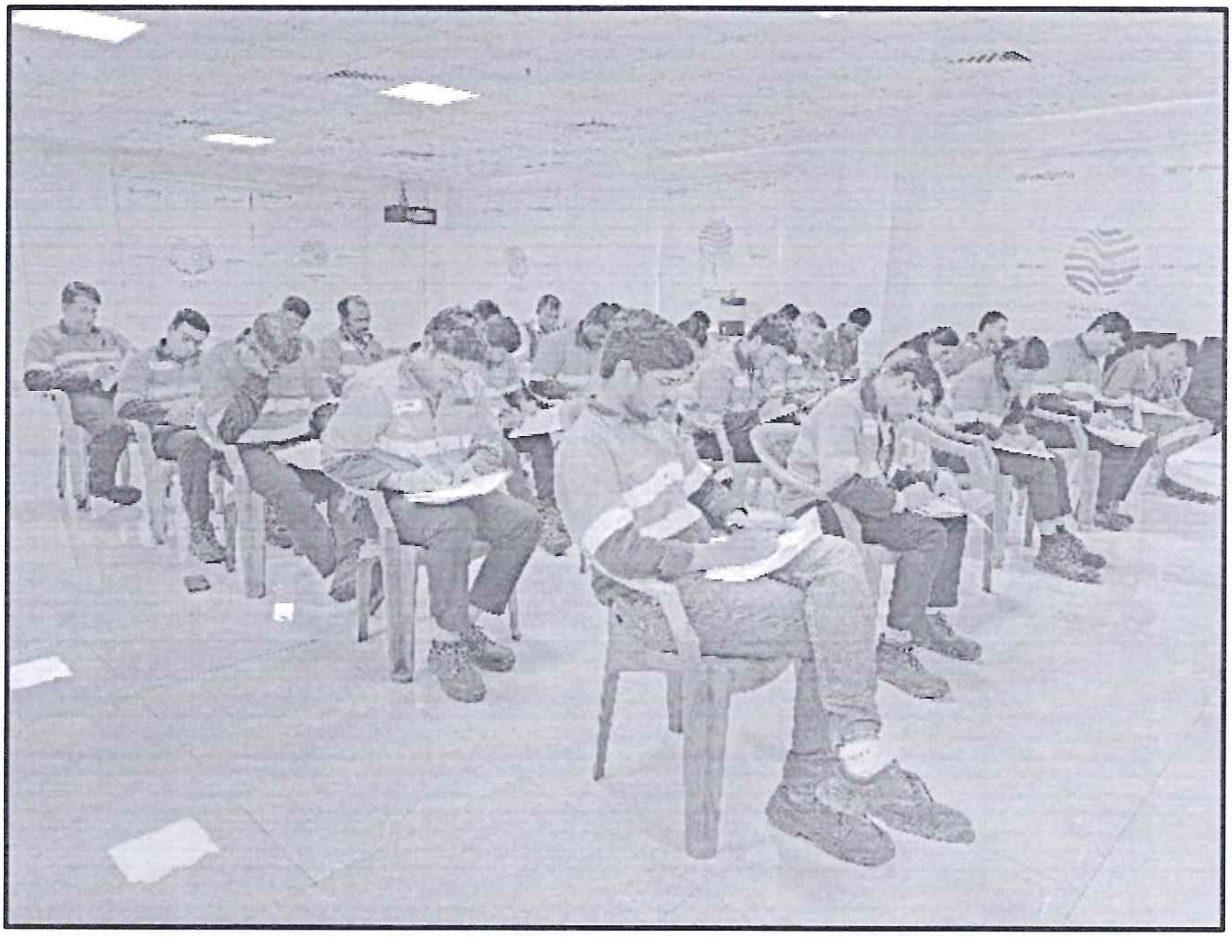
- Since time is the essence, the Occupational health specialist shall monitor the victim and start the revival process by initiating CPR while travelling to the nearest tie-up hospital.
- On the way, provide adequate oxygen supply with help of O2 cylinder with nose mask.
- Monitor the blood SPo2 levels continuously.
- If required provide electrical shock with Automatic external Defibrillator (AED) if pulse drops considerably.

Training:

Training both classroom and onsite plays a very important role in building competency among the rescue team to face any eventuality.

Steps involved are:

- Form a rescue team with involvement of employees and business partners.
- Competency mapping and gap analysis of the rescue team
- Provide them with adequate rescue equipment.
- Provide training on Confined space rescue.
- Conduct on-site training on rescue procedures.
- Check the areas of improvement and re-train.
- Identify various scenarios of confined space and plan for emergency drills.
- Conduct planned and surprise drill and check its effectiveness.
- Reward and recognize best performing rescuers (individual) and team.



Summary:

In every consideration TIME is the critical factor and rescue should be done as quickly as possible, but 100% safe for the rescue team.
The speed with which the system can be deployed, and the rescue carried out is vitally important, as is the SIMPLICITY and EASE of use so that a typical employee can deploy and carry out a rescue after being trained.

Whichever methodology is chosen, the target time should be to rescue the suspended employee in under five minutes.

This is a dynamic document that shall be reviewed, audited, and updated:

- At least annually
- As required due to a rescue process change or implemented.
- As the result of lessons learned from testing the plan or having to utilize the plan to perform a rescue.

Format no. __________

Confined space rescue Plan & monitoring effectiveness

Confined space work Dates: From: .... To: ....

What task(s) to be performed? Check all that apply:

- ☐ Repair and refurbishment of refractory lining
- ☐ Acid Cleaning of tank
- ☐ Maintenance work
- ☐ Others ____________________

Names of employees who are engaged for Confined space work:

Printed Name	Signature
1. ____________	____________
2. ____________	____________
3. ____________	____________
4. ____________	____________
5. ____________	____________
6. ____________	____________
7. ____________	____________
8. ____________	____________
9. ____________	____________
10. ____________	____________
11. ____________	____________
12. ____________	____________

Does the task require a rescue Plan?

a. Yes　　　　　　b. No

If yes, name and designation of the rescuers

1. ____________________ (Team leader)
2. ____________________ (Authorized Gas tester)
3. ____________________
4. ____________________
5. ____________________
6. ____________________
7. ____________________
8. ____________________
9. ____________________
10. ____________________

Person Responsible for Confined Space Rescue (CSR)

Name	Signature
____________________	____________________

Name (Backup)	Signature
____________________	____________________

Very Important- All the persons engaged in Confined space will be responsible for their own health and safety.

Communication:

What communication systems will be used between the members of the rescue team and supervisor / rescue team lead?

- ☐ Direct voice communication
- ☐ Whistle
- ☐ Mobile Phone
- ☐ Two-way Radios / Headsets

Emergency Contact:

In the event of an emergency/ Asphyxiation, the attendant shall alert the Confined Space Rescue (CSR) team leader, who in turn shall immediately instruct:

1) Emergency Management Team (EMT)/ Emergency Response Team (ERT)
2) HSE Manager
3) Fire Department

Rescue team review:

2) Are employees involved in the rescue process competent in the use of rescue equipment like SCBA, PFAS, Tripod Ascend system, motorized winch, first aid measure?

   a. Yes　　　　b. No

3) Are rescue training records up to date?

   a. Yes　　　　b. No

4) Is there enough rescuers available?

   a. Yes　　　　b. No　　　　If yes, Number?

5) Is rescue equipment selected appropriate for the nature of work?

   a. Yes　　　　b. No

6) What obstructions are in the way of reaching the suspended employee?

   a. ______________________________

7) What is the plan to overcome the obstruction?

   a. ______________________________

8) Have assessments been made of anchor points?

   a. Yes　　　　b. No

9) What rescue equipment is required & will be used to rescue the suspended employee? Check all that apply.

- ☐ Motorized Winch lifting capacity 500 kgs.
- ☐ Tripod of height 3 meters with retractable system and Winch
- ☐ Rescue Kit (300 kgs. Capacity) with 50-meter kernmantle rope
- ☐ Ascend- Descent Rescue Kit (300 kgs. Capacity) with 100-meter kernmantle
- ☐ Cantilever arm with powered winch of lifting capacity 500 kgs.
- ☐ Climbing Rope Rescue System of length 10 meters and 20 meters
- ☐ Full Body harness with double lanyard and scaffold hook for each rescuer
- ☐ Vertical stretcher/s

Testing the Plan

1) Has the plan been tested?

   a. Yes   b. No

   b. When?

   ____

2) Please describe the rescue process.

   a. Rescue technique used-

      - ☐ Self-Rescue
      - ☐ Assisted Self Rescue
      - ☐ Rescue with Ascend rescue system.

3) How long did it take to complete the rescue?

   Time (In minutes)- ______________________________

   Any concerns- ______________________________

Protecting the Scene

How will the scene be protected from change prior to completing a full assessment of the root cause?

- ☐ Move non-essential personnel to an Assembly point.
- ☐ Keep non-essential personnel out of the area by setting up a barricade.
- ☐ Begin assessing the scene immediately after the suspended person has been treated and moved from the area.
- ☐ Move the asphyxiated person to tie-up hospital in Ambulance along with OH specialist.
- ☐ Preserve wreckage.
- ☐ Report the incident immediately to the proper channels.
- ☐ Obtain approval to photograph the scene.

Effectiveness check: Rate them 0 to 5 (Where 0 is lowest & 5 being highest)

- ☐ Awareness level of Rescue team ()
- ☐ Use of appropriate rescue procedure ()
- ☐ Rescue was executed within timelines ()
- ☐ Support team like maintenance, HSE and OHC executed their role effectively ()

Opportunities for improvement: To be recommended by HSE Personnel

Note: Other than this job specific rescue plan made for each confined space entry procedure equipment wise. This Rescue procedure SC/SVS/CSRP-01 can be referred for detailed and different scenarios for rescue.